INCH-POUND

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MILITARY SPECIFICATION

BOOTS, COLD WEATHER, INSULATED, RUBBER

(WET-COLD; DRY-EXTREME COLD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers both black and white insulated rubber boots. The black boot is for wear in cold, wet climates. The white boot is for wear in extreme cold, dry climates.

1.2 Classification. The boots shall be of the following types, classes, and sizes, as specified (see 6.2).

Type I	Boot, wet-cold (black)
Class 1	- Rigid sole
Class 2	- Flexible sole
Class 3	- Flexible sole with safety toe
Type II	Boot, dry-extreme cold (white)
Class 1	- Deleted (see 6.9)
Class 2	- Vapor barrier, polyurethane injected sole

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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TABLE I. Schedule of sizes

Width	3	4	5	6	7	8	9	10	11	12	13	14
XN	-	X	X	X	X	X	X	X	X	X	X	X
N	X	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203 - Paper, Kraft, Untreated
C-F-206 - Felt Sheet: Cloth, Felt, Wool, Pressed
V-L-61 - Laces, Nylon
CCC-C-429 - Cloth, Osnaburg, Cotton
CCC-C-446 - Cloth, Muslin, Cotton
MMM-A-188 - Adhesive; Urea-Resin-Type (Liquid and Powder)
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-C-7219 - Cloth, Duck, Nylon
MIL-C-17238 - Cloth, Knitted, Wool and Cotton Fleece
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-284 - Visual Inspection Guide for Rubber Footwear
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF COMMERCE

Commercial Standard 35-61 - Hardwood Plywood

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

C 177 - Steady-State Thermal Transmission Properties by Means of the Guarded Hot Plate
D 297 - Method for Rubber Products - Chemical Analysis
D 412 - Rubber Properties in Tension
D 573 - Rubber-Deterioration in an Air Oven
D 746 - Test Method for Brittleness Temperature of Plastics and Elastomers by Impact
D 792 - Specific Gravity and Density of Plastics by Displacement
D 1044 - Initial Tear Resistance of Plastic Film and Sheeting
D 1630 - Rubber Property-Abrasion Resistance (NBS Abrader)
D 2240 - Rubber Property-Durometer Hardness
D 3951 - Standard Practice for Commercial Packaging

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D 4066 - Nylon Injection and Extrusion Materials
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z41.1 American National Standard for Men's Safety Toe Footwear

(Copies may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Material. All cloth to be coated except the felt component (see 3.3.6) shall contain not more than 0.003 percent copper when tested as specified in 4.4.1.1. All cloth to be coated except the felt component (see 3.3.6) and the nylon tricot (see 3.3.3) shall contain not more than 0.0015 percent manganese when tested as specified in 4.4.1. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Cloth, muslin, cotton. The cotton muslin cloth shall conform to type I, class 1 of CCC-C-446 except that the requirements for shrinkage shall not apply.

3.3.1.1 Cloth, bootleg duck. Bootleg duck shall be a minimum of 7.2 ounces per square yard, nominal 26 by 21, plain weave, coating quality, cotton.

3.3.2 Cloth, osnaburg, cotton. All cotton osnaburg cloth shall conform to type II, class 3 or 5 of CCC-C-429. On the type I boots the osnaburg cloth for the eyelet stay form and the outer eyelet stay shall be dyed black.

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3.3.3 Cloth, nylon tricot. The nylon tricot shall be made from continuous filament 40 ± 2 denier, 13 filament, bright yarn. The color shall be natural (undyed). The finished cloth shall be a 2-bar jersey tricot construction conforming to the requirements of table II when tested as specified in 4.4.1.1.

TABLE II. Requirements for nylon tricot

	Minimum	Maximum
Weight oz. per sq. yd.	3.3	--
Wales per inch	50	60
Courses per inch	50	60
Bursting strength, pounds	90.0	--
Elongation coursewise, percent	120	150

3.3.3.1 Finish (nylon tricot cloth). The cloth shall be thoroughly scoured and heat set. Both vertical edges of each piece of cloth shall be gummed with liquid nylon, or equal, 5/8 ± 1/8 inch in width.

3.3.4 Cloth, nylon, knitted. The circular knitted nylon cloth shall be made from continuous filament 70 ± 2 denier, 34 filament, bright yarn. The color shall be black for the type I boot lace chafing strip and hinge upper stay, and natural for the type II boot lace chafing strip, hinge upper stay, and all nylon inner linings. The finished cloth shall be circular knitted with two ends of the 70 ± 2 denier yarn knitting as a single end and shall conform to the requirements of table III when tested as specified in 4.4.1.1.

TABLE III. Requirements for knitted nylon cloth

Knitted stitch	1 by 1 rib (double thread rib)
Weight, oz. per sq. yd., minimum	3.8
Wales per inch, minimum	21
Courses per inch, minimum	30
Bursting strength, pounds, minimum	125

3.3.4.1 Finish (nylon knitted cloth). The cloth shall be thoroughly scoured, split and finished to the full open flat width.

3.3.5 Cloth, knitted, wool and cotton fleece. The fleece cloth shall conform to the requirements of MIL-C-17238.

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3.3.5.1 Insulation, foam. Insulation for the type II, class 2 boot shall be composed of polyurethane foam, 1.55 pounds per cubic foot, needlepunched on one side with a 3 and 4 denier polyester fiber and coated on the other side with a polyurethane film 1 mil thick. The composite shall meet the requirements of table IV.

TABLE IV. Requirements for foam insulation

Weight (composite)	10.42 ($\pm$ 0.50) oz/yd^2
Thickness - uncompressed	7/64 $\pm$ 1/64 inch
- compressed	5/64 $\pm$ 1/32 inch
Thermal resistance (R value) (BTU/HR FT^2 degrees F)	0.9646
Needlepunched fiber	Polyester
Coating on foam side	Polyurethane

3.3.6 Felt, pressed. The felt used in the boots shall be type I, classification number 9R3 for the toe cap and type I, classification number 12R3 for the midsole, conforming to C-F-206 (see 3.6.13 and 3.6.14 for thickness required).

3.3.7 Cloth, nylon twill. The cloth for the inner eyelet stays shall be either a 2/2 twill made of high tenacity, continuous filament, 210 $\pm$ 2 denier, bright yarn, conforming to the requirements of table V, when tested as specified in 4.4.1.1 or a nylon duck cloth conforming to type III, class I of MIL-C-7219. The color shall be white or natural.

TABLE V. Requirements for nylon woven cloth

	Minimum	Maximum
Weight, oz. per sq. yd.	4.70	5.10
Ends per inch	73	79
Picks per inch	74	80
Breaking strength, pounds:		
Warp	325	---
Filling	330	---

3.3.8 Rubber and polyurethane compounds.

3.3.8.1 Upper rubber compounds. The rubber latex and calender compounds for the uppers shall conform to the requirements in table VI when tested as specified in 4.4.1.1. After accelerated aging in an air oven for 16 hours minimum at 212 $\pm$ 2 degrees F, each upper compound shall retain not less than 75 percent of its original tensile strength and ultimate elongation when tested as specified in 4.4.1.1. The upper compounds shall meet the requirements for brittleness when tested as specified in 4.4.1.1.

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TABLE VI. Upper rubber compound requirements

	Color	Tensile strength, minimum, psi	Ultimate elongation, percent (min)
Type I, classes 1, 2 and 3			
Latex compound	Black 1/	2500	500
Calendered compound	Black 1/	1700	450
Type II			
Latex compound	White	2500	500
Calendered compound	White	1500	450

1/ Except that the toe cap color shall be white (see 3.3.15).

3.3.8.2 Outsole and heel rubber compounds. The outsole and heel rubber compounds shall conform to the requirements in table VII when tested as specified in 4.4.1.1. After accelerated aging in an air oven for 16 hours minimum at 212 ± 2 degrees F, each compound shall have an abrasive index of not less than 75 percent of its abrasive index before aging when tested as specified in 4.4.1.1. Each compound shall meet the requirements for brittleness when tested as specified in 4.4.1.1.

TABLE VII. Outsole and heel rubber compound requirements

	Color	Specific gravity, maximum	Abrasive index, minimum	Hardness (units)
Type I, class 1:				
Outsole	Black	1.2	80	60 ± 5
Heel	Black	1.2	80	70 ± 5
Type I, classes 2 and 3	Buff	1.1	50	45 ± 5
Type II	White	1.2	50	45 ± 5

3.3.8.3 Midsole, outsole, and heel polyurethane components. The polyurethane components shall conform to the requirements in table VIII when tested as specified in 4.4.1.1. After accelerated aging for 7 days at 70 degrees C and 100 percent relative humidity and then conditioning for 24 hours at 25 degrees C and 50 percent relative humidity, each component shall retain at least 80 percent of its original tensile strength when tested as specified in 4.4.1.1.

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TABLE VIII. Polyurethane components requirements

Polyurethane Midsole:	
Density	25 to 31 pounds per cubic foot
Hardness, Shore A	40 to 45
Tear strength	85 to 95 pounds
Tensile strength	550 to 650 psi
Elongation	390 to 430 percent
Modulus at -30°C of the expanded material and the solid material	14,000 psi, maximum
Ross-flex, 1/4 inch panel, (at -20°F)	0 percent cut growth at 10,000 cycles
Polyurethane outsole and heel:	
Density	62.4 pounds per cubic foot, minimum
Hardness, Shore A	52 to 57
Tear strength	180 pounds, minimum
Tensile strength	1200 to 1600 psi
Elongation	450 to 650 percent
Ross-flex (at -20°F)	Not less than 900 percent cut growth at 50,000 cycles

3.3.8.4 Ski strap shelf compound. The rubber compound for the ski strap shelf shall have a hardness requirement of 60 ± 5 units when tested as specified in 4.4.1.1.

3.3.9 Wood shank unit. The shank length and shank size schedule shall be as specified in tables IX and X. The wood shank shall be of plywood molded at the time of gluing to fit the profile of the last bottom from heel to designated ball area. The plywood unit shall be constructed from three plies of veneer. The core, or inside ply, shall run at right angle to the outer plies and shall extend to between 7/8 and 1 inch from the front edge. A fourth ply of mahogany shall be inserted in the curve portion of the shank between the top veneer and the core. The grain direction of the mahogany shall be parallel to the veneer core. The three plies shall be birch veneer with a minimum of 1/24 inch thickness. The plies shall be grade 1, except that the core ply may be grade 2, as defined in Commercial Standard 35-61. The face and back veneers shall have smooth surfaces. The assembled shank thickness shall be 0.100 to 0.125 inch when measured in the heel area, 1/2 inch from the rear edge. The mahogany insert shall extend approximately 3 inches from a point 1/8 inch to 1/16 inch from the edge of the shank. The front edge of the shank shall consist of only the two outer plies and shall be feathered. When strips of veneer are edge joined to form sheets, the strips shall preferably be tapeless spliced. However, if tape-splicing is used, the veneers shall be smoothly jointed so that the resulting joint will be tight with no openings or gaps. After the unit is completed, tape shall be removed without damage to the wood surfaces. The

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plywood shall be glued with an adhesive conforming to MMM-A-188. The adhesive shall be applied in an adequate and uniform manner as recommended by the adhesive manufacturer. Assembly periods, and clamping or pressing times and pressures shall be in accordance with the glue manufacturer's instructions in order to insure a tight uniform bond with no voids. When the shanks are tested as specified in 4.4.1.1, there shall be a minimum of 20 percent wood failure in each area tested.

3.3.9.1 Alternate nylon shank unit. As an alternate, a nylon shank may be used in lieu of the wood shank. The nylon shank shall be made from nylon conforming to ASTM D 4066, group 1, class 1, grade IV and shall have a thickness of 0.125 ± 0.0125 inch when tested as specified in 4.4.1.1. The shank length and shank size schedule shall be as specified in tables IX and X.

TABLE IX. Shank length

Shank size No.	Shank length 1/ inches	Shank size No.	Shank length 1/ inches	Shank size No.	Shank length 1/ inches	Shank size No.	Shank length 1/ inches
3	5-1/4	6	5-13/16	9	6-3/8	12	6-15/16
4	5-7/16	7	6	10	6-9/16	13	7-1/8
5	5-5/8	8	6-3/16	11	6-3/4	14	7-5/16

1/ A tolerance of ± 1/16 inch is permitted.

TABLE X. Shank size schedule

	Boot Size											
WIDTH	3	4	5	6	7	8	9	10	11	12	13	14
XN	-	4	4	5	6	7	8	9	10	11	12	13
N	3	4	5	6	7	8	9	10	11	12	13	14
R	3	4	5	6	7	8	9	10	11	12	13	14
W	3	4	5	6	7	8	9	10	11	12	13	14
XW	3	4	5	6	7	8	9	10	11	12	13	14

3.3.10 Laces. The laces shall conform to type II, class 3 of V-L-61. The color of the laces shall be black for type I boots, and white for type II boots. The length of the laces shall be a minimum of 60 inches.

3.3.11 Eyelets. The eyelets shall be of telescopic type made of aluminum sheet not less than 0.015 inch in thickness, with a roll setting barrel, and shall conform to the requirements shown in table XI when tested as specified in 4.4.1.1.

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TABLE XI. Eyelet requirements

Measurement	Minimum inches	Maximum inches
Outside diameter of flange	0.535	0.545
Outside diameter of barrel	0.300	0.305
Diameter of hole	0.220	0.230
Overall length	0.195	0.205

3.3.11.1 Finish of eyelets. After fabrication, the eyelets shall be cleaned and tumble-coated with not less than two coats of hard baked enamel. The color shall be black for type I boots and white for type II boots.

3.3.11.2 Washers. The washers shall be made of aluminum sheet 0.013 ± 0.0015 inch thick, have a scored barrel, and conform to the finished requirements shown in table XII when tested as specified in 4.4.1.1.

TABLE XII. Washer requirements

Measurement	Minimum inches	Maximum inches
Outside diameter of flange	0.535	0.545
Diameter of hole	0.283	0.288
Overall length	0.135	0.145

3.3.12 Valve. The valve shall be Schrader Automotive Products Valve No. 9917 or Bata Shoe Company Valve No. 89141, or equal (see 6.8). Prior to the use of the "or equal" item, the contractor shall furnish a sample of the valve, with supporting data to the contracting officer for subsequent evaluation by the responsible Military agency. The valve shall be a manually operated air release valve making a water and air tight seal when closed.

3.3.13 Instruction tag. Instruction tags shall be fabricated using tagstock and printed in black ink. The tag shall be a 6-1/2 inches by 5-1/2 inches tagstock sheet folded lengthwise to form a four-page booklet having two outer printed pages and two inner printed pages. The tags shall be perforated in the upper left hand corner to provide for insertion of lace end. The printing shall start on page number one and read from top to bottom. The instruction tag for the type I boots shall be printed on blue tinted tagstock and contain the printed matter shown on figure 4. The instruction tag for the type II boots shall be printed on white tagstock and contain the printed matter shown on figure 5. Size of characters shall be 1/16 inch high minimum except capital letters shall be 3/32 inch high minimum.

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3.3.14 Steel box toe. The steel box toe for the type I, class 3 boots shall conform to the toe of the last and shall be fabricated from cold-rolled carbon steel, with a Rockwell C hardness of 43 to 50, carbon content of 0.50 and 0.82 and thickness of 0.0625, ±0.0025 inch when tested as specified in 4.4.1.1. The steel box toe shall be thoroughly cleaned and completely coated with a zinc compound or any other suitable coating that will retard corrosion and not damage adjacent parts of the boot (see 3.9.1).

3.3.15 Bumper toe cap. The bumper toe cap for the type I, class 3 boots shall be made of rubber upper compound conforming to 3.3.8.1, except the color shall be white.

3.3.16 Adhesive. Unless otherwise specified herein, the following types of adhesives shall be used for joining the various components of the boot. The specific type used for any specific cementing operation will be chosen by the contractor. The type chosen shall provide a secure bond between the joined components. The adhesive shall be applied in accordance with the manufacturer's instructions.

a. Natural rubber latex
b. Synthetic rubber latex (including chloroprene)
c. Natural rubber solvent cement
d. Synthetic rubber solvent cement
e. Synthetic resin cement

3.4 Lasts and patterns.

3.4.1 Lasts. The lasts shall be provided by the contractor and approved by the contracting activity, and shall conform to the requirements as shown in tables XIII through XVIII. (See figure 2). The surface of the last shall be smooth, clean, and free from any protrusions due to faulty casting.

3.4.1.1 Last measurements. The tolerance for the finished bottom length shall be ± 1 point; and for ball, waist, choke, ankle, and leg girth shall be ± 2 points. Finished last bottoms shall conform to measurements of bottom patterns furnished by the Government (see 6.5), with tolerances of ± 1/32 inch in length and ± 1/48 inch in width. All dimensions, except tolerances for bottom length and width, are to be taken by means of sticks and tapes whose calibrations are geometrically graded and which shall be loaned by the Government.

3.4.2 Models, patterns and templates. The last models, patterns and templates will be furnished by the Government (see 6.5). The patterns and dies shall be provided by the contractor.

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TABLE XIII. Last dimensions in geometric points and inch fractions for XN width

Measurement 1/	Geom. dif.	Size 4	5	6	7	8	9
Bottom length	10	1068	1078	1088	1098	1108	1118
		9-15/24	9-23/24	10-6/24	10-13/24	10-21/24	11-5/24
Bottom width	10	683	693	703	713	723	733
(ball)		3-2/24	3-4/24	3-6/24	3-8/24	3-10/24	3-12/24
Ball girth	10	980	990	1000	1010	1020	1030
		7-25/64	7-38/64	7-54/64	8-6/64	8-22/64	8-38/64
Waist girth	10	975	985	995	1005	1015	1025
		7-18/64	7-33/64	7-48/64	7-63/64	8-14/64	8-30/64
Choke girth	8	1174	1182	1190	1198	1206	1214
		13-14/64	13-35/64	13-56/64	14-14/64	14-36/64	14-58/64
Ankle girth	6	1102	1108	1114	1120	1126	1132
		10-42/64	10-54/64	11-3/64	11-16/64	11-29/64	11-42/64
Leg girth "H"	6	1129	1135	1141	1147	1153	1159
		11-36/64	11-49/64	11-63/64	12-13/64	12-27/64	12-42/64
Leg Girth "J"	6	1179	1185	1191	1197	1203	1209
		13-27/64	13-45/64	13-59/64	14-11/64	14-28/64	14-44/64
Leg girth "K"	6	1211	1217	1223	1229	1235	1241
		14-49/64	15-2/64	15-20/64	15-38/64	15-56/64	16-10/64

1/ See figure 2.

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TABLE XIII. Last dimensions in geometric points and inch fractions for XN width (cont'd)

Measurement 1/	Geom. dif.	Size 10	11	12	13	14
Bottom length	10	1128	1138	1148	1158	1168
		11-13/24	11-21/24	12-6/24	12-15/24	13
Bottom width (ball)	10	743	753	763	773	783
		3-15/24	3-18/24	3-21/24	4	4-3/24
Ball girth	10	1040	1050	1060	1070	1080
		8-55/64	9-8/64	9-26/64	9-44/64	9-63/64
Waist girth	10	1035	1045	1055	1065	1075
		8-47/64	9	9-17/64	9-35/64	9-54/64
Choke girth	8	1222	1230	1238	1246	1254
		15-17/64	15-41/64	16-1/64	16-26/64	16-52/64
Ankle girth	6	1138	1144	1150	1156	1162
		11-56/64	12-6/64	12-20/64	12-34/64	12-49/64
Leg girth "H	6	1165	1171	1177	1183	1189
		12-56/64	13-6/64	13-22/64	13-38/64	13-56/64
Leg girth "J"	6	1221	1227	1233	1239	1245
		14-61/64	15-13/64	15-31/64	15-49/64	16-4/64
Leg girth "K"	6	1247	1253	1259	1265	1271
		16-29/64	16-48/64	17-4/64	17-24/64	17-44/64

1/ See figure 2.

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TABLE XIV. Last dimensions in geometric points and inch fractions for N width

Measurement 1/	Geom. dif.	Size 3	4	5	6	7	8	9
Bottom length	10	1060	1070	1080	1090	1100	1110	1120
		9-10/24	9-17/24	10	10-7/24	10-14/24	10-22/24	11-6/24
Bottom width	10	683	693	703	713	723	733	743
		3-1/24	3-4/24	3-6/24	3-8/24	3-10/24	3-12/24	3-15/24
Ball girth	10	990	1000	1010	1020	1030	1040	1050
		7-38/64	7-54/64	8-6/64	8-22/64	8-38/64	8-55/64	9-8/64
Waist girth	10	985	995	1005	1015	1025	1035	1045
		7-33/64	7-48/64	7-63/64	8-14/64	8-30/64	8-47/64	9
Choke girth	8	1174	1182	1190	1198	1206	1214	1222
		13-14/64	13-35/64	13-50/64	14-13/64	14-36/64	14-58/64	15-17/64
Ankle girth	6	1102	1108	1114	1120	1126	1132	1138
		10-42/64	10-54/64	11-3/64	11-16/64	11-29/64	11-42/64	11-56/64
Leg girth "H"	6	1129	1135	1141	1147	1153	1159	1165
		11-36/64	11-49/64	11-63/64	12-13/64	12-27/64	12-42/64	12-56/64
Leg girth "J"	6	1179	1185	1191	1197	1203	1209	1215
		13-27/64	13-45/64	13-59/64	14-11/64	14-28/64	14-44/64	14-61/64
Leg girth "K"	6	1211	1217	1223	1229	1235	1241	1247
		14-50/64	15-2/64	15-20/64	15-38/64	15-56/64	16-10/64	16-29/64

1/ See figure 2.

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TABLE XIV. Last dimensions in geometric points and inch fractions N width (cont'd)

Measurement 1/	Geom. dif.	Size 10	11	12	13	14
Bottom length	10	1130	1140	1150	1160	1170
		11-14/24	11-23/24	12-8/24	12-17/24	13-2/24
Bottom width	10	753	763	773	783	793
(ball)		3-18/24	3-21/24	4	4-3/24	4-6/24
Ball girth	10	1060	1070	1080	1090	1100
		9-26/64	9-44/64	9-63/64	10-18/64	10-38/64
Waist girth	10	1055	1065	1075	1085	1095
		9-17/64	9-35/64	9-54/64	10-9/64	10-28/64
Choke girth	8	1230	1238	1246	1254	1262
		15-41/64	16-1/64	16-25/64	16-51/64	17-13/64
Ankle girth	6	1144	1150	1156	1162	1168
		12-6/64	12-20/64	12-34/64	12-49/64	13
Leg girth "H"	6	1171	1177	1183	1189	1195
		13-7/64	13-22/64	13-38/64	13-56/64	14-6/64
Leg girth "J"	6	1221	1227	1233	1239	1245
		15-14/64	15-31/64	15-49/64	16-4/64	16-23/64
Leg girth "K"	6	1253	1259	1265	1271	1277
		16-48/64	17-4/64	17-24/64	17-44/64	18

1/ See figure 2.

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TABLE XV. Last dimensions in geometric points and inch fractions for R width

Measurement 1/	Geom. dif.	Size 3	4	5	6	7	8	9
Bottom length	10	1062	1072	1082	1092	1102	1112	1122
		9-11/24	9-18/24	10-2/64	10-8/24	10-16/24	11	11-8/24
Bottom width	10	693	703	713	723	733	743	753
(ball)		3-4/24	3-6/24	3-8/24	3-10/24	3-12/24	3-15/24	3-18/24
Ball girth	10	1010	1020	1030	1040	1050	1060	1070
		8-6/64	8-22/64	8-38/64	8-55/64	9-8/64	9-26/64	9-44/64
Waist girth	10	1005	1015	1025	1035	1045	1055	1065
		7-62/64	8-14/64	8-30/64	8-47/64	9	9-17/64	9-35/64
Choke girth	8	1182	1190	1198	1206	1214	1222	1230
		13-35/64	13-50/64	14-13/64	14-36/64	14-58/64	15-17/64	15-41/64
Ankle girth	6	1108	1114	1120	1126	1132	1138	1144
		10-55/64	11-3/64	11-16/64	11-29/64	11-42/64	11-56/64	12-6/64
Leg girth "H"	6	1135	1141	1147	1153	1159	1165	1171
		11-49/64	11-63/64	12-13/64	12-27/64	12-42/64	12-56/64	13-7/64
Leg girth "J"	6	1185	1191	1197	1203	1209	1215	1221
		13-51/64	13-59/64	14-11/64	14-28/64	14-44/64	14-61/64	15-14/64
Leg girth "K"	6	1217	1223	1229	1235	1241	1247	1253
		15-2/64	15-20/64	15-38/64	15-56/64	16-10/64	16-29/64	16-48/64

1/ See figure 2.

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TABLE XV. Last dimensions in geometric points and inch fractions for R width (cont'd)

Measurement 1/	Geom. dif.	Size				
		10	11	12	13	14
Bottom length	10	1132	1142	1152	1162	1172
		11-16/24	12-1/24	12-10/24	12-19/24	13-4/24
Bottom width	10	763	773	783	793	803
(ball)		3-21/24	4	4-3/24	4-6/24	4-9/24
Ball girth	10	1080	1090	1100	1110	1120
		9-63/64	10-18/64	10-38/64	10-59/64	11-16/24
Waist girth	10	1075	1085	1095	1105	1115
		9-54/64	10-9/64	10-28/64	10-49/64	11-6/64
Choke girth	8	1238	1246	1254	1262	1270
		16-5/64	16-25/64	16-51/64	17-13/64	17-40/64
Ankle girth	6	1150	1156	1162	1168	1174
		12-20/64	12-24/64	12-49/64	13	13-15/64
Leg girth "H"	6	1177	1183	1189	1195	1201
		13-22/64	13-38/64	13-54/64	14-6/64	14-22/64
Leg girth "J"	6	1227	1233	1239	1245	1251
		15-31/64	15-49/64	16-4/64	16-23/64	16-42/64
Leg girth "K"	6	1259	1265	1271	1277	1283
		17-4/64	17-24/64	17-44/64	18	18-21/64

1/ See figure 2.

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TABLE XVI. Last dimensions in geometric points and inch fractions for W width

Measurement 1/	Geom. dif.	Size 3	4	5	6	7	8	9
Bottom length	10	1064	1074	1084	1094	1104	1114	1124
		9-12/24	9-20/24	10-3/24	10-10/24	10-18/24	11-2/24	11-10/24
Bottom width	10	703	713	723	733	743	753	763
(ball)		3-6/24	3-8/24	3-10/24	3-12/24	3-15/24	3-18/24	3-21/24
Ball girth	10	1030	1040	1050	1060	1070	1080	1090
		8-38/64	8-55/64	9-8/64	9-29/64	9-44/64	9-63/64	10-18/64
Waist girth	10	1025	1035	1045	1055	1065	1075	1085
		8-30/64	8-47/64	9	9-17/64	9-34/64	9-54/64	10-9/64
Choke girth	8	1190	1198	1206	1214	1222	1230	1238
		13-21/64	14-13/64	14-36/64	14-58/64	15-17/64	15-41/64	16-1/64
Ankle girth	6	1114	1120	1126	1132	1138	1144	1150
		11-3/64	11-16/64	11-29/64	11-42/64	11-56/64	12-6/64	12-20/64
Leg girth "H"	6	1141	1147	1153	1159	1165	1171	1177
		11-63/64	12-13/64	12-27/64	12-42/64	12-56/64	13-7/64	13-22/64
Leg girth "J"	6	1191	1197	1203	1209	1215	1221	1227
		13-59/64	14-11/64	14-28/64	14-44/64	14-61/64	15-14/64	15-31/64
Leg girth "K"	6	1223	1229	1235	1241	1247	1253	1259
		15-20/64	15-38/64	15-56/64	16-10/64	16-29/64	16-48/64	17-4/64

1/ See figure 2.

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TABLE XVI. Last dimensions in geometric points and inch fractions for W width (cont'd)

Measurement 1/	Geom. dif.	Size 10	11	12	13	14
Bottom length	10	1134	1144	1154	1164	1174
		11-18/24	12-3/24	12-12/24	12-21/24	13-6/24
Bottom width	10	773	783	793	803	813
(ball)		4	4-3/24	4-6/24	4-9/24	4-12/24
Ball girth	10	1100	1110	1120	1130	1140
		10-38/64	10-59/64	11-16/64	11-38/64	11-61/64
Waist girth	10	1095	1105	1115	1125	1135
		10-28/64	10-49/64	11-9/64	11-27/64	11-49/64
Choke girth	8	1246	1254	1262	1270	1278
		16-26/64	16-52/64	17-13/64	17-40/64	18-3/64
Ankle girth	6	1156	1162	1168	1174	1180
		12-34/64	12-49/64	13	13-15/64	13-30/64
Leg girth "H"	6	1183	1189	1195	1201	1207
		13-38/64	13-54/64	14-6/64	14-22/64	14-38/64
Leg girth "J"	6	1233	1239	1245	1251	1257
		15-49/64	16-4/64	16-23/64	16-42/64	16-61/64
Leg girth "K"	6	1265	1271	1277	1233	1289
		17-24/64	17-44/64	18	18-21/64	18-42/64

1/ See figure 2.

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TABLE XVII. Last dimensions in geometric points and inch fractions for XW width

Measurement 1/	Geom. dif.	Size 3	4	5	6	7	8	9
Bottom length	10	1066	1076	1086	1096	1106	1116	1126
		9-14/24	9-22/24	10-4/24	10-12/24	10-19/24	11-3/24	11-11/24
Bottom width	10	713	723	733	743	753	763	773
(ball)		3-8/24	3-10/24	3-12/24	3-15/24	3-18/24	3-21/24	4
Ball girth	10	1050	1060	1070	1080	1090	1100	1110
		9-8/64	9-26/64	9-44/64	9-63/64	10-18/64	10-38/64	10-59/64
Waist girth	10	1045	1055	1065	1075	1085	1095	1105
		8-63/64	9-17/64	9-35/64	9-54/64	10-9/64	10-28/64	10-49/64
Choke girth	8	1198	1206	1214	1222	1230	1238	1246
		13-14/64	14-36/64	14-58/64	15-17/64	15-41/64	16-1/64	16-25/64
Ankle girth	6	1120	1126	1132	1138	1144	1150	1156
		11-16/64	11-29/64	11-42/64	11-56/64	12-6/64	12-20/64	12-24/64
Leg girth "H"	6	1147	1153	1159	1165	1171	1177	1183
		12-12/64	12-27/64	12-42/64	12-56/64	13-7/64	13-22/64	13-38/64
Leg girth "J"	6	1197	1203	1209	1215	1221	1227	1233
		14-11/64	14-28/64	14-44/64	14-61/64	15-14/64	15-31/64	15-49/64
Leg girth "K"	6	1229	1235	1241	1247	1253	1259	1265
		15-38/64	15-56/64	16-10/64	16-29/64	16-48/64	17-4/64	17-24/64

1/ See figure 2.

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TABLE XVII. Last dimensions in geometric points and inch fractions for XW width (cont'd)

Measurement 1/	Geom. dif.	Size 10	11	12	13	14
Bottom length	10	1136	1146	1156	1166	1176
		11-20/24	12-4/24	12-13/24	12-22/24	13-8/24
Bottom width	10	783	793	803	813	823
(ball)		4-3/24	4-6/24	4-9/24	4-12/24	4-14/24
Waist girth	10	1115	1125	1135	1145	1155
		11-7/64	11-28/64	11-49/64	12-8/64	12-32/64
Choke girth	8	1254	1262	1270	1278	1286
		16-51/64	17-13/64	17-40/64	18-3/64	18-32/64
Ankle girth	6	1162	1168	1174	1180	1186
		12-49/64	13	13-15/64	13-30/64	13-45/64
Leg girth "H"	6	1189	1195	1201	1207	1213
		13-56/64	14-6/64	14-22/64	14-38/64	14-56/64
Leg girth "J"	6	1239	1245	1251	1257	1263
		16-4/64	16-23/64	16-42/64	16-61/64	17-17/64
Leg girth "K"	6	1271	1277	1283	1289	1295
		17-44/64	18	18-21/64	18-42/64	18-63/64

1/ See figure 2.

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TABLE XVIII. Location of "H", "J", "K" measurements of last

Measurement 1/	Size						
	3	4	5	6	7	8	9
"H" Position	844	850	856	862	869	875	881
	4-9/10	5	5-1/10	5-2/10	5-3/10	5-4/10	5-5/10
"J" Position	981	985	989	994	998	1002	1007
	7-4/10	7-5/10	7-6/10	7-7/10	7-8/10	7-9/10	8
"K" Position	1062	1065	1068	1071	1074	1078	1081
	9-4/10	9-5/10	9-6/10	9-7/10	9-8/10	9-9/10	10

TABLE XVIII. Location of "H", "J", "K" measurements of last (cont'd)

Measurement 1/	Size				
	10	11	12	13	14
"H" Position	887	893	899	904	910
	5-6/10	5-7/10	5-8/10	5-9/10	6
"J" Position	1011	1015	1019	1023	1027
	8-1/10	8-2/10	8-3/10	8-4/10	8-5/10
"K" Position	1084	1088	1091	1094	1097
	10-1/10	10-2/10	10-3/10	10-4/10	10-5/10

1/ Location of positions for "H", "J", "K" measurements (see figure 2) are expressed in geometric points and inch fractions for each size. Above measurements for heights, predicated on a perpendicular last position using a 7/8 inch heel, taken at location of breast of heel from bottom of last upward.

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3.5 Design.

3.5.1 Type I (black boot). The type I boots shall be black, blucher type, with eyelets and laces for ankle and leg adjustments. The class 1 boots shall have a black, rigid non-slip type outer sole and molded heel, a toe cap and an air release valve (see figure 1). The class 2 boots shall have an air release valve, a buff-color, flexible non-slip type outer sole and molded heel (see figure 3). The class 3 boot shall be similar to class 2, except it shall have an insulated protective steel box toe positioned between the inner vamp and the outer vamp insulation, and a white bumper toe cap. All type I boots shall be vulcanized as a unit consisting of four layers of materials as follows:

(a) An impermeable seamless inner lining sealer.
(b) An inner lining of fabric cemented to inner lining sealer.
(c) An outer lining of fabric separated by an air space from the inner lining but adhered with cement to the upper.
(d) An impermeable seamless outer surface (upper).

The insulated area shall consist of the inner vamp, outer vamp and heel linings, and the felt midsole; and shall measure 6-1/2 ± 1/2 inches in height from the top of the insole section of the inner lining face on a line perpendicular to the heel breast area.

3.5.2 Type II (white boot). The type II boots shall be white, blucher type, rigid sole with eyelets and laces for ankle and leg adjustments (see figure 6), and have a toe cap and an air release valve. The color of the boots shall approximately match color No. 37778 of FED-STD-595. The outer sole and heel shall be white, non-slip type with a molded heel. Boots shall be vulcanized as a unit consisting of five layers of materials as follows:

(a) An impermeable seamless inner lining sealer.
(b) An inner lining of fabric cemented to inner lining sealer.
(c) An interlining of fabric.
(d) An outer lining of fabric separated by an air space from the inner lining but adhered with cement to the upper.
(e) An impermeable seamless outer surface (upper).

The insulation shall consist of the inner and outer linings, the interlinings, and the felt midsole; and shall cover the area of boot indicated by measurements specified in table XIX.

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TABLE XIX. Insulation requirements (type II boot)

Area	Insulating components
0 to 3 inches 1/	No insulation
3 to 4 inches 1/	One layer (outer lining)
4 inches and below 1/	Three layers (inner lining, interlining and outerlining)
Bottom 2/	Felt midsole

1/ Measurements taken from top of boot.

2/ The felt midsole shall be used as insulation for the bottom of the boot, and additional insulation of the foot section shall be supplied by the interlining.

3.6 Construction. All operations shall be performed as stated below. Each part shall be substantially rolled or pressed and all critical junctures shall be edged. Unless otherwise specified, all measurements stated are based on size 9R. All knitted fabric components shall be cut with the wales running in the long direction.

3.6.1 Inner lining sealer. The inner lining sealer shall be seamless and consist of natural rubber latex which shall closely conform to the last. The overall thickness of the inner lining sealer shall be 0.018 inch at a point 6 inches from the top of the boot.

3.6.2 Nylon inner lining. The nylon inner lining shall be made of natural color knitted nylon cloth conforming to 3.3.4, spread with cement, coated with rubber coating compound on same side to an overall thickness of 0.026 inch, and shall be adhered with cement to the inside surface of the inner lining seal, so as to extend to within 2 ± 1/2 inches from the top of boot at center back to within 3 inches from the top of boot at center front.

3.6.3 Inner lining seal strip. The inner lining seal strip shall be made of muslin cloth conforming to 3.3.1, frictioned on two sides and coated on two sides with rubber coating compound to an overall thickness of 0.020 inch, and adhered with cement to the coated side of the nylon inner lining at the seam to form a seal strip.

3.6.4 Inside friction vamp. An inside friction vamp of osnaburg cloth, conforming to 3.3.2, frictioned on two sides and coated on one side with rubber coating compound to an overall thickness of 0.023 inch, shall be adhered with cement between inner lining sealer and the inner lining to act as a reinforcement.

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3.6.5 Inside toe stiffener. An inside toe stiffener of osnaburg cloth conforming to 3.3.2, frictioned on two sides and coated on one side with a rag compound to an overall thickness of 0.040 inch, shall be adhered with cement to the outer side of the inside friction vamp and the toe to stiffen the toe portion. The inside toe stiffener shall not be applied to type I, class 3 boots having steel safety toes.

3.6.6 Steel toe, type I, class 3 boot. The type I, class 3 boot shall have the steel toe specified in 3.3.14 placed over the inner friction vamp. A 1-inch piece of cotton osnaburg cloth conforming to 3.3.2, frictioned on two sides and coated on one side with rubber coating compound to an overall thickness of 0.023 inch, shall be adhered with cement over the top edge of the steel toe with one half adhering to the steel and the remainder adhering to the vamp of the boot. The toe shall be completely insulated and the osnaburg cloth shall completely cover the entire steel toe cap.

3.6.7 Friction filler. A friction filler of osnaburg cloth conforming to 3.3.2, frictioned on two sides and coated on one side with a good grade of stiffening compound to an overall thickness of 0.040 inch, shall be adhered with cement to the outside of the inner lining sealer to reinforce the inner lining sealer across the entire sole portion.

3.6.8 Heel stiffener. An inside counter of a compound of fiber and rubber or a good grade of stiffening compound treated to give a hardness of 80 to 90 when tested as specified in 4.4.1.1 and having an overall thickness of 0.080 inch, shall be adhered with cement to the inner lining sealer at the counter section and shall be skived on all margins. The heel stiffener shall be not more than 1-1/2 inches high on a finished size 9R boot inside measurement, and shall grade not more than 1/16 inch in height per size. The heel stiffener shall be long enough to extend not less than 3/4 inch beyond the breast of the heel.

3.6.9 Heel stiffener (alternate). As an alternate, the inside counter shall consist of three laminated layers each of osnaburg cloth conforming to 3.3.2, and shall be frictioned on two sides and coated on one side with a rag stiffener compound to an overall thickness of 0.035 inch. The counter shall be skived all around 1/4 to 3/8 inch in width. The remaining requirements shall be as specified in 3.6.8.

3.6.10 Inside friction counter. An inside counter of osnaburg cloth conforming to 3.3.2, frictioned on both sides and coated on one side with rubber coating compound to an overall thickness of 0.023 inch, shall cover the heel stiffener. The lower margin of the inside friction counter shall extend not less than 1 inch below the bottom surface of the heel section of the felt midsole to anchor the inner lining and felt midsole together.

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3.6.11 Inner vamp lining. The inner vamp lining shall consist of two half sections of fleece cloth conforming to 3.3.5 or a single section skived at the top. The knitted back of the fleece cloth shall be spread with cement, coated with 0.008 inch of rubber coating compound, adhered with cement to the inner lining sealer and the inside friction vamp, and extend to the bottom edge of the applied felt midsole. The inner vamp lining of the type II, class 2 boot shall meet the requirements of 3.3.5.1. It shall be cemented on the film side and not rubber coated.

3.6.12 Inner quarter lining. The inner quarter lining shall be of one piece of fleece cloth conforming to 3.3.5 and skived on the top edge. The knitted back of the fleece shall be spread with cement, coated with 0.008 inch of rubber coating compound, adhered with cement to the inner lining sealer reinforcement and inside friction counter, and extend to the bottom edge of the applied felt midsole. The inner quarter lining of the type II, class 2 boot shall meet the requirements of 3.3.5.1. It shall be cemented on the film side and not rubber coated.

3.6.13 Inside toe cap. For type I, class 1 boots only, a shaped or seamed toe cap, 6 $\pm$ 1/8 by 3 $\pm$ 1/8 inches of 1/8 inch felt conforming to 3.3.6, shall be placed at the toe over inner vamp lining. The bottom edge of this toe cap shall be cemented, and adhered to the bottom edge of the applied felt midsole. The inside toe cap of the type II, class 2 boot shall meet the requirements of 3.3.5.1. It shall be cemented on the film side and not rubber coated.

3.6.14 Midsole shank unit. The midsole shall consist of a shank, in accordance with 3.3.9 or 3.3.9.1, sandwiched in between two plies of 5/16 inch pressed wool felt for type I boots conforming to 3.3.6. Both sides of the shank unit shall be bonded with cement to the felt. The shank unit shall extend from the heel to designated ball area. The felt shall be coated on both sides with cement and bonded together and further bonded on one side to the inside friction filler, and on the other side to the friction sole form. The felt midsole shall conform to the dimensions and pattern of the last bottom after the seamless lining face and the nylon inner lining have been applied. The type II, class 2 boot shall use a midsole of one layer of 5/16 inch pressed wool felt with the shank between the felt and midsole.

3.6.15 Interlining. For type II boots only, the interlining shall extend to the bottom edge of the applied midsole. A two-piece interlining of knitted fleece cloth conforming to 3.3.5 shall be positioned between the inner and outer lining assemblies over the vamp and heel sections. The fleece cloth shall be spread with cement and coated with 0.008 inch of rubber coating compound on one side. The napped side of the cloth shall be placed toward the last. The interlining of the type II, class 2 boot shall meet the requirements of 3.3.5.1. It shall be cemented on the film side and not rubber coated.

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3.6.16 Lasting strip. For type II boots only, a lasting strip of muslin cloth conforming to 3.3.1, frictioned on two sides and coated with rubber coating compound on one-side to an overall thickness of 0.020 inch, shall be used at the bottom margins of the interlining to adhere the interlining to the felt midsole.

3.6.16.1 Interlining front and back seam strips. For type II boots only, the interlining front and back seam strips shall be 5/8 inch wide cut strips of muslin cloth conforming to 3.3.1, frictioned on two sides and coated on one side with rubber coating compound to an overall thickness of 0.020 inch and shall be used to join the two halves of the interlining.

3.6.17 Outer vamp lining. The outer vamp lining shall consist of one piece of knitted fleece cloth conforming to 3.3.5. The knitted back of the fleece cloth shall be spread with cement and coated with 0.008 inch of rubber coating compound and adhered with cement to the vamp sealer; so placed that when adhered to the boot, the lower margin of the outer vamp lining shall overlap the lower edge of the midsole and the top margin butt join (or slightly overlap) the "waist" edge of the outer quarter lining. The outer vamp lining of the type II, class 2 boot shall meet the requirements of 3.3.5.1. It shall be cemented on the film side and not rubber coated.

3.6.18 Outer quarter lining. The outer quarter lining shall be of one piece of knitted fleece cloth conforming to 3.3.5. The knitted back of the fleece cloth shall be spread with cement, coated with 0.008 inch of rubber coating compound, and adhered with cement to the boot by the quarter sealer. The top edge shall be skived and cemented on the fleece side with the cemented surface adhered to the inner lining. The outer quarter lining shall overlap the bottom edge of the applied midsole. The outer quarter lining of the type II, class 2 boot shall meet the requirements of 3.3.5.1. It shall be cemented on the film side and not rubber coated.

3.6.19 Vamp sealer. The outer vamp lining shall be reinforced and sealed on its outer margins by the vamp sealer of nylon tricot conforming to 3.3.3, calender coated with rubber coating compound on one side to a thickness of 0.010 inch and coated on the reverse side with 0.006 inch of rubber coating compound (or spread with cement on both sides and coated with 0.008 inch of rubber coating compound on one side). The sealer shall be similar in shape to the outer vamp lining, except there shall be sufficient margin at the top edge to overlap and seal onto the quarter sealer and sufficient margin on the bottom edge to form a lasting margin under the midsole. The cement coated side is to be bonded to the latex upper.

3.6.20 Quarter sealer. The outer quarter lining shall be reinforced and sealed on its outer margins by the quarter sealer made of nylon tricot conforming to 3.3.3, calender coated with rubber coating compound on one side to a thickness of 0.010 inch and coated on the reverse side with 0.006 inch of rubber coating compound (or spread with cement on both sides and coated with 0.008 inch of rubber coating compound on one side). The quarter sealer shall be similar in shape to the outer quarter lining except that its margin shall extend

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as follows: at the top, to the full extent of the boot; on the bottom, to form the lasting margin; to the midsole and at the front, to overlap beyond the one edge of the outer quarter lining to seal seam, but not to extend beyond the outer quarter lining at the "butt" joint of the outer vamp lining at the "waist". The cement coated side is to be bonded to the latex upper with cement.

3.6.20.1 Vamp and quarter joints sealer strip. The vamp and quarter sealer strip shall be of upper compound conforming to 3.3.8.1, 0.020 to 0.030 inch thick, 1-1/2 ± 1/16 inches wide. The sealer strip shall be adhered to the vamp and quarter sealer joints in order to prevent leakage of the joint. The sealer strip shall be not less than 1/2 inch under the boot bottom and shall extend across the vamp and quarter sealer joint, equally spaced to opposite side of the boot, not less than 1/2 inch under the boot bottom.

3.6.21 Friction sole form. A friction sole form of osnaburg cloth conforming to 3.3.2, frictioned on two sides and coated on one side with rubber coating compound to an overall thickness of 0.023 inch shall cover the bottom surface of the felt midsole with its margins extending sufficiently beyond the midsole to overlap the lasting strip.

3.6.22 Combination vamp and chafing piece. A combination vamp and chafing piece shall reinforce the vamp and lower lacing area. This combination vamp and chafing piece shall be of nylon tricot conforming to 3.3.3, calender-coated with rubber coating compound on one side to a thickness of 0.010 inch and coated on the reverse side with 0.006 inch of rubber coating compound (or spread with cement on both sides and coated with 0.008 inch of rubber coating compound on one side). The pattern of this piece shall be cut right and left, and placed in the boot with the extended instep section sloping upwards into the medial or inner leg. To save unnecessary weight, the lower margins of the vamp area do not extend to the bottom of the boot but are so designed that there will be an overlap of no more than 1/4 inch by the friction foxing. The cement coated side is to be bonded to the latex upper with cement.

3.6.23 Upper. A seamless upper, of rubber latex conforming to 3.3.8.1, 0.018 inch in thickness, 6 inches from the top, shall cover and be adhered to the entire surface of the outside lining and any exposed friction or reinforcing. The upper shall be assembled to the boot prior to the attachment of the outside parts covered by 3.6.24 through 3.6.40.

3.6.24 Heel stay. A heel stay of osnaburg conforming to 3.3.2, frictioned two sides and coated one side with a compound of fiber and rubber to an overall thickness of 0.040 inch, shall be adhered with cement between the upper and the friction heel piece.

3.6.25 Friction toe foxing. A friction toe foxing of osnaburg cloth conforming to 3.3.2, 2 inches wide, frictioned on both sides and coated with a rubber coating compound on one side to an overall thickness of 0.023 inch, shall extend around the forepart of the boot from the heel breast to heel breast and be adhered with cement. The type II, class 2 boot shall not have a friction toe foxing.

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3.6.26 Toe foxing. A toe foxing, 2-1/8 inches wide and 0.028 to 0.039 inch thick of upper compound conforming to 3.3.8.1, shall cover the frictioned toe foxing and shall be adhered with cement. The top margin shall overlap onto the upper not less than 1/8 inch. The type II, class 2 boot shall not have a toe foxing.

3.6.27 Friction heel piece. A heel piece of osnaburg cloth conforming to 3.3.2, frictioned on two sides and coated on one side with rubber coating compound to an overall thickness of 0.023 inch, shall be adhered with cement to the upper and the heel stay. The type II, class 2 boot shall not have a friction heel piece.

3.6.28 Heel piece. A heel piece of upper compound conforming to 3.3.8.1, 0.028 to 0.039 inch thick, shall cover the frictioned heel piece over-lapping onto the upper by 3/8 inch and be adhered with cement. The type II, class 2 boot shall not have a heel piece.

3.6.29 Lace chafing piece. The lace chafing piece, made of knitted nylon cloth conforming to 3.3.4, spread with cement, and coated with rubber coated compound on same side to an overall thickness of 0.026 inch, shall be adhered with cement to the instep with the extended section sloping upwards into the medial or inner leg.

3.6.30 Eyelet stays. Eyelet stays, four to a pair of boots, shall be placed on the boot. Each assembled eyelet stay shall consist of an eyelet stay form, inner eyelet stay, outer eyelet stay, eyelet stay cover, lower hinge stay, and upper hinge stay. At least a 1/2 inch margin of the eyelet stay shall be adhered with cement along the rear margin of the eyelet stay to the boot upper.

3.6.30.1 Eyelet stay cover. The eyelet stay cover shall be of upper compound conforming to 3.3.8.1, 0.028 to 0.039 inch thick. The margins shall be a minimum of 3/32 inch wide except that front edge at hinge stay may taper to 0.

3.6.30.2 Eyelet stay form. The eyelet stay form shall be of osnaburg cloth conforming to 3.3.2, frictioned both sides and coated one side with rubber coating compound to an overall thickness of 0.23 inch.

3.6.30.3 Outer eyelet stay. The outer eyelet stay shall be of osnaburg cloth conforming to 3.3.2, frictioned both sides and coated one side with rubber coating compound to an overall thickness of 0.23 inch.

3.6.30.4 Inner eyelet stay. The inner eyelet stay shall be of nylon cloth conforming to 3.3.7, frictioned, spread or otherwise impregnated on both sides and coated on one side with rubber coating compound to an overall thickness of 0.23 inch.

3.6.30.5 Lower hinge stay. A lower hinge stay of upper compound conforming to 3.3.8.1, 0.039 inch thick shall form an anchor between the inner eyelet stay and the latex upper at the blucher point.

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3.6.30.6 Upper hinge stay. The upper hinge stay piece, made of knitted nylon cloth conforming to 3.3.4, spread with cement, and coated with rubber coating compound on same side to an overall thickness of 0.026 inch, shall be adhered with cement to the inside surface of the outer and inner eyelet stay at the top of the eyelet stay and extending below the first two eyelets, and shall be hinged to the boot gusset upper 3/4 ± 1/16 inch.

3.6.31 Toe cap. For type I, class 1 and 3, a toe cap of upper compound conforming to 3.3.8.1, 0.032 inch thick, 3 ± 1/8 inches wide and 6 ± 1/8 inches long, shall cover the toe surface of the upper.

3.6.32 Outside sole form. An outside sole form of osnaburg cloth conforming to 3.3.2, frictioned on both sides with rubber coating compound, shall be adhered with cement between the bottom of the boot and the outsole.

3.6.32.1 Type II, class 2. The outside sole form on the type II, class 2 boot shall be made from bootleg duck, frictioned and rubber coated on one side. This form shall extend around the bottom of the boot and up the sides to a line 1/8 to 3/8 inch above the nip line of the polyurethane injection mold.

3.6.32.2 Heel filler. A heel filler of 0.031 inch thick of gum rubber shall be adhered to the osnaburg cloth in the heel area to provide a level surface for the heel.

3.6.33 Outsole. The outsole shall be fabricated from the rubber compound conforming to 3.3.8.2, except that the sole and heel combination for the type II, class 2 boot shall be a two density injection molded sole conforming to 3.3.8.3.

3.6.33.1 Type I, class 1 (groove-grip chevron outsole). The outsole shall be of a groove-grip chevron type design (see figure 1). The dimensions of the outsole at the ball shall be 0.300 inch thick at the chevron bars, 0.160 inch between the chevron bars, and shall be 0.100 inch thick at the heel seat. The shank shall be tapered in thickness from 0.175 inch back of the tread to 0.100 inch at the heel base. The color of the outsole shall be black.

3.6.33.2 Type I, class 2 and class 3 (traction outsole). The outsole shall be of a suitable design (see figure 3) for traction purposes, and shall be 0.185 inch thick at the ball section and 0.125 inch at the shank and under the heel. The color of the outsole shall be buff.

3.6.33.3 Type II, class 2 (polyurethane outsole/heel). The injected polyurethane heel and sole unit shall have a chevron pattern as shown in figure 6. The dimension between the chevron bars shall be 0.025 inches. The thickness at the ball shall be 1 inch, 7/16 inch at the shank, and 1-7/16 inch at the heel. The color shall be white, approximating that of the rubber components. The sole/heel combination shall be graded to be compatible with the outer configuration of the assembled upper. The sole assembly shall be direct injected as a dual density unit on fully vulcanized upper assemblies.

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3.6.34 Heel. A molded 7/8 inch rubber heel conforming to 3.3.8.2 and of an approved design shall be attached to the boot. The width and length shall be as specified in table XX. The contractor shall submit a sample of his proposed heel design to the contracting officer for approval prior to commencing production.

3.6.34.1 Type I, class 1. The color shall be black.

3.6.34.2 Type I, classes 2 and 3. The color shall be buff.

3.6.34.3 Type II, class 2. The heel shall be integral with the sole. The pattern shall be as defined in figure 6.

TABLE XX. Heel measurements, inches 1/

Boot-size	Heel measurement	Boot-width XN	N	R	W	XW
3	Width	-	2-5/8	2-5/8	2-3/4	2-3/4
	Length		3	3	3-1/8	3-1/8
4 & 5	Width	2-5/8	2-3/4	2-3/4	2-7/8	2-7/8
	Length	3	3-1/8	3-1/8	3-1/4	3-1/4
6	Width	2-3/4	2-3/4	2-7/8	2-7/8	3
	Length	3-1/8	3-1/8	3-1/4	3-1/4	3-3/8
7	Width	2-3/4	2-7/8	2-7/8	3	3
	Length	3-1/8	3-1/4	3-1/4	3-3/8	3-3/8
8	Width	2-7/8	2-7/8	3	3	3-1/8
	Length	3-1/4	3-1/4	3-3/8	3-3/8	3-1/2
9	Width	2-7/8	3	3	3-1/8	3-1/8
	Length	3-1/4	3-3/8	3-3/8	3-1/2	3-1/2
10	Width	3	3	3-1/8	3-1/8	3-1/4
	Length	3-3/8	3-3/8	3-1/2	3-1/2	3-5/8
11	Width	3	3-1/8	3-1/8	3-1/4	3-1/4
	Length	3-3/8	3-1/2	3-1/2	3-5/8	3-5/8
12	Width	3-1/8	3-1/8	3-1/4	3-1/4	3-3/8
	Length	3-1/2	3-1/2	3-5/8	3-5/8	3-3/4
13	Width	3-1/8	3-1/4	3-1/4	3-3/8	3-3/8
	Length	3-1/2	3-5/8	3-5/8	3-3/4	3-3/4
14	Width	3-1/4	3-1/4	3-3/8	3-3/8	3-1/2
	Length	3-5/8	3-5/8	3-3/4	3-3/4	3-7/8

1/ A tolerance of ± 1/16 inch is permitted for all dimensions.

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3.6.35 Outsole wedge. To attain flatness of tread, a wedge of outsole rubber compound filling stock conforming to 3.3.8.2, triangular in shape 3/8 inch on base and 0.160 inch thick, shall be adhered with cement to the outer margin of the fore part of the outsole from shank to shank, except that the type II, class 2 boot shall not use an outsole bind strip..

3.6.36 Outsole bind strip. An outsole bind strip of outsole rubber conforming to 3.3.8.2, 1/2 inch wide, 0.100 inch thick at its lower edge, and 0.035 inch thick at its upper edge shall be positioned and adhered with cement around the outsole edge from shank to shank so as to fully cover and secure the outsole wedge. The type II, class 2 boot shall not use an outsole bind strip.

3.6.37 Ski strap shelf. A ski strap shelf of rubber compound shall be firmly attached with cement (centered) above the heel with the lower edge contacting the outsole. The dimensions of the shelf shall be as shown on figures 1 and 3.

3.6.38 Valves. For type I, class 1, 2 and 3, a manually-opened air release valve conforming to 3.3.12 shall be positioned in the center of the outside quarter not less than 6 inches from the base of the heel. Valves are to be in a closed position when boots are packed.

3.6.38.1 Valve stay. For type I, class 1, 2 and 3 boots, a valve stay of upper rubber compound conforming to 3.3.8.1, 1-3/4 -0, +1/8 inch in diameter and 0.028 to 0.039 inch thick, shall seal the valve on the outer surface of the valve base using cement.

3.6.38.2 Valve base anchor. For type I, class 1, 2 and 3 boots, a valve base anchor of upper rubber compound conforming to 3.3.8.1, 0.025 to 0.035 inch thick may be placed under the valve to insure adhesion between the base of the valve and the latex upper. The valve assembly components shall be cemented and pressed together mechanically to create a permanent airtight seal.

3.6.39 Finish of boot. For type I, classes 2 and 3 boots, the finish shall be bright and lacquered glossy smooth.

3.6.40 Eyeletting. There shall be six eyelets placed on each side of the boot, uniformly spaced 3/8 $\pm$ 1/8 inch from the front edge of the boot to the edge of the eyelet. The eyelets shall be reinforced with washers. The washers shall be on the inside surface of the boot. All eyelets shall be firmly clinched so that the setting is smooth and free from breaks and cuts. The eyelet holes may be cut prior to vulcanization and the eyelets placed after vulcanization.

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3.6.41 Lacing. The laces shall be single laced through the bottom eyelet, double laced through the second eyelet, and single laced through the third eyelet. A natural or buff pressboard form shall be placed in the leg of the boot and the laces tightly drawn and tied to crease the gusset area against the form.

3.6.42 Height of boot. The height of the boot shall be as specified herein. The boot shall be measured from the bottom of the heel (inside breast) to the top of the boot.

Type I, class 1	-	Overall outside height shall be 11-1/4 ± 1/4 inches.
Type I, classes 2 and 3	-	Overall outside height shall be 12 ± 1/4 inches.
Type II, class 2	-	Overall outside height shall be 11-1/4 ± 1/4 inches.

3.6.43 Pairing. The boots shall be paired and tied firmly together by the ends of the laces.

3.6.44 Marking and identification. The manufacturer's code letters and the month and year (expressed in numerals) of the date of the contract in 1/8 ± 1/16 inch characters, together with the letters "US" and the size and width, in 5/16 ± 1/16 inch characters, shall be permanently embossed or laid in characters using rubber ink on a rubber unsupported label or labels and placed on the boot at the top of the gum heel piece. The rubber labels for type I boots shall have a black background with white or yellow letters and numerals; type II boots, a white background with black letters and numerals. As an alternate, a white rubber under a black rubber stenciled overlay may be used for type I boots, and a black rubber under a white rubber stenciled overlay may be used for type II boots. The item nomenclature and contract number shall be stamped with black indelible marking ink on the nylon inner lining in the upper quarter area using characters 1/8 inch high minimum. The marking shall be clearly legible. A size marking may be applied to the shank area of the outsole.

Example of rubber label: XY
US
3-90
9R

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3.6.44.1 Special marking. The following shall be permanently and clearly imprinted with yellow letters for type I boots and with black letters for type II boots (see figure 1 and figure 3):

(a) On the valve seal for type I class 1, 2 and 3, and type II class 2: "KEEP CLOSED UNLESS AIRBORNE" and "OPEN" (1/8 to 1/4 inch letters) and two arrows as shown on figure 1.

(b) On the lateral eyelet stay directly behind the three top eyelets: "FOLD GUSSET THIS SIDE ONLY" (1/4 to 3/8 inch letters).

(c) On the lateral eyelet stay next to bottom eyelet: "KEEP DOUBLELACED TO HOLD FIRMLY" (1/4 to 3/8 inch letters).

3.6.44.2 ANSI labeling. The type I, class 3 boots shall be labeled to indicate that they conform to class 75 safety toe requirements outlined in the American National Standards Institute ANSI Z41.1. ANSI marking of the nylon inner lining on upper quarter shall be in accordance with the nomenclature and contract labeling requirements of 3.6.44 and shall be separated from the nomenclature and contract information. The label inscription and format shall be as follows:

ANSI Z41 PT 83
M 1/75 C/75

3.6.45 Instruction tag. One instruction tag as specified in 3.3.13 shall be tied to each pair of boots using the end of a lace.

3.6.46 Weight of boot. Prior to the leakage test, each finished boot shall be weighed without laces, instruction tag, and boot form, to the nearest 0.5 ounce. This weight shall be branded on the inner side of the heel of each boot. Figures shall be 3/8 $\pm$ 1/8 inch high and shall have a minimum depth of 1/32 inch.

3.6.47 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in table XXII shall be removed from production and replaced with nondefective and properly matched components.

3.7 Tolerances. For all rubber compounds and fabrics, unless otherwise specified, the tolerance for all weights, dimensions, and thicknesses shall be plus or minus 10 percent. The tolerance is not applicable to the finished boot weight.

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3.8 Vulcanization. The boots shall be fully vulcanized under pressure so that all rubber parts will be cured into a single unit, that is, the layers of rubber used in fabrication of the boot shall not be separable into their component parts at their original interface, in accordance with normal commercial vulcanization procedures. There shall be no blisters or air pockets in the seams.

3.8.1 Bleeder holes. Any bleeder holes on type I, class 1, 2 and 3 boots required to bleed off air during vulcanization, shall be plugged with self curing cement which shall be forced into the holes.

3.9 Performance requirements.

3.9.1 Impact resistance, type I, class 3 boot. The steel box toe of the type I, class 3 boot shall have a minimum inside clearance of 1/2 inch when tested for impact resistance as specified in 4.4.7.

3.9.2 Leakage. After vulcanization, the boot shall not leak when tested for leakage as specified in 4.4.2.1 and 4.4.7.

3.9.3 Adhesion of components. After vulcanization, there shall be no evidence of separation (delamination) or looseness of any of the boot components when tested as specified in 4.4.2.2.

3.10 Workmanship. The end item shall conform to the quality of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in table XXII, the dimensions specified in table XX and 3.6.4.2, and tested as specified in 4.4.7.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing. Testing shall be performed on components and materials listed in table XXI. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table XXI, they shall be reported as required by the referenced test methods. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the lot average fail to meet any requirements specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 to and including 22,000	3
22,001 and over	5

For each type of rubber compound, the sample unit shall be a representative composite of the lot of rubber prepared in the form indicated below. A lot of rubber compound other than latex (separate lot for each type) shall consist of the amount of rubber compound that is produced or compounded in one day. A lot of latex compound shall consist of no more than the amount of latex used for one week's production of boots. The composite test sample of rubber shall be vul-

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canized in the same manner employed in the manufacture of the boots and the test unit (sample unit) shall consist of the number of pieces formed to the dimensions as follows:

(a) Upper latex compound - two pieces 6 by 6 inches by 0.025 +0.010, -0.0 inch and one piece 6 by 6 inches by 0.060 +0.010, -0.0 inch to be used in the low temperature flexibility test.

(b) Upper calender compound - two pieces 6 by 6 inches by 0.070 ± 0.010 inch.

(c) Outsole compound - two pieces 6 by 6 inches by 0.070 ± 0.010 inch and one piece 6 by 6 inches by 0.250 ± 0.025 inch.

(d) Heel compound - two pieces 6 by 6 inches by 0.070 ± 0.010 inch and one piece 6 by 6 inches by 0.250 ± 0.025 inch.

(e) Ski strap shelf - one piece 6 by 6 inches by 0.250 ± 0.25 inch.

(f) Stiffening compound - one piece 6 by 6 inches by 0.250 ± 0.025 inch.

(g) Polyurethane compounds - two pieces 6 by 6 inches by 0.250 ± 0.025 inches.

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TABLE XXI. Component tests

Component and lot size expressed in terms of of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Nylon yarn (1 cone, spool, or tube)	Denier	3.3.3, 3.3.4, 3.3.7	1/	-	-	-	-	-	-
	No. of filaments	3.3.3, 3.3.4	1/	-	-	-	-	-	-
	Lustre	3.3.3, 3.3.4, 3.3.7	Visual 1/	-	-	-	-	-	-
	Tenacity	3.3.7	1/	-	-	-	-	-	-
Cloth, cotton, muslin, 1 yard full width	Copper and manganese	3.3	1/	-	-	-	-	-	-
Cloth, cotton, osnaburg, 1 yard full width	Copper, and manganese	3.3	1/	-	-	-	-	-	-
Cloth, nylon tricot, 1 yard full width	Mat'l identification	3.3.3	1/	-	-	-	-	-	
	Color	3.3.3	Visual 1/	-	-	-	-	-	
	Weight	3.3.3	5041 2/	X	-	-	-	-	
	Wales and courses	3.3.3	5070 2/	X	-	-	-	-	1 yard full width
	Bursting strength	3.3.3	5120 2/	X	-	-	-	-	
	Elongation (coursewise)	3.3.3	4.5.1	X	-	5	1.0 percent	-	
	Finish	3.3.3.1	Visual 1/	-	-	-	-	-	
	Copper	3.3	1/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Cloth, knitted nylon, 1 yard full width	Mat'l identification	3.3.4	1/	-	-	-	-	-	
	Color	3.3.4	Visual 1/	-	-	-	-	-	
	Knitted stitch	3.3.4	Visual 1/	-	-	-	-	-	1 yard full
	Weight	3.3.4	5041 2/	X	-	-	-	-	width
	Wales per courses	3.3.4	5070 2/	X	-	-	-	-	
	Bursting strength	3.3.4	5120 2/	X	-	-	-	-	
	Finish	3.3.4.1	Visual 1/	-	-	-	-	-	
	Copper and manganese	3.3	1/	-	-	-	-	-	
Cloth, knitted, wool and cotton fleece	Copper and manganese	3.3	1/	-	-	-	-	-	-
Foam insulation	Weight	3.3.5.1	5041 2/	-	-	-	-	-	
	Compressed thickness	3.3.5.1	5030 2/	-	-	-	-	-	
	Uncompressed thickness	3.3.5.1	Visual 1/	-	-	-	-	-	
	Heat transfer	3.3.5.1	C 177-76 3/	-	-	-	-	-	
Cloth, nylon twill, 1 yard full width	Mat'l identification	3.3.7	1/	-	-	-	-	-	
	Weave	3.3.7	Visual 1/	-	-	-	-	-	
	Weight	3.3.7	5041 2/	X	-	-	-	-	1 yard
	Ends of picks	3.3.7	1/	-	-	-	-	-	full width
	Breaking strength	3.3.7	5100 2/	X	-	-	-	-	
	Copper and manganese	3.3	1/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Upper compounds latex and calender (batch)	Tensile strength	3.3.8.1	D 412 (method A) 3/	X	-	-	-	-	
	Elongation	3.3.8.1	D 412 (method A) 3/	X	-	-	-	-	
	Tensile strength after accelerated aging	3.3.8.1	7001, D 573 3/ and D 412 (method A) 3/	X	-	-	-	-	
	Elongation after accelerated aging	3.3.8.1	7001, D 573 3/ and D 412 (method A) 3/	X	-	-	-	-	See 4.4.1
	Brittleness temperature	3.3.8.1	D 746	X	-	-	-	-	
Outsole and heel compounds (batch)	Specific gravity	3.3.8.2	D 792 3/	X	-	-	-	-	
	Abrasive index (before aging)	3.3.8.2	D 1630 3/	X	-	-	-	-	
	Abrasive index (after aging)	3.3.8.2	D 1630 3/	X	-	-	-	-	See 4.4.1
	Hardness	3.3.8.2	D 2240 3/	X	-	-	-	-	
	Brittleness temperature	3.3.8.1	D 746	X	-	-	-	-	
Polyurethane midsole	Density	3.3.8.3	D 297						
	Hardness	3.3.8.3	D 2240 3/	-	-	-	-	-	
	Tear strength	3.3.8.3	D 1004 3/	-	-	-	-	-	
	Tensile strength	3.3.8.3	D 412 3/	-	-	-	-	-	
	Elongation	3.3.8.3	D 412 3/	-	-	-	-	-	
	Accelerated aging	3.3.8.3	D 412 3/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Polyurethane midsole (cont'd)	Modulus	3.3.8.3	D 790 3/	-	-	-	-	-	
	Ross-flex	3.3.8.3	D 1052 3/	-	-	-	-	-	
Polyurethane outsole and heel	Density	3.3.8.3	D 297						
	Hardness	3.3.8.3	D 2240 3/	-	-	-	-	-	
	Tear strength	3.3.8.3	D 1004 3/	-	-	-	-	-	
	Tensile strength	3.3.8.3	D 412 3/	-	-	-	-	-	
	Elongation	3.3.8.3	D 412 3/	-	-	-	-	-	
	Accelerated aging	3.3.8.3	D 412 3/	-	-	-	-	-	
	Ross-flex	3.3.8.3	D 1052 3/	-	-	-	-	-	
Ski strap shelf compound (batch)	Hardness	3.3.8.4	D 2240 3/	X	-	-	-	-	See 4.4.1
Stiffening compounds (batch)	Hardness	3.6.8.	D 2240 3/	X	-	-	-	-	See 4.4.1
Wood shank unit (1 gross pairs)	Mat'l identification	3.3.9	1/	-	-	-	-	-	
	Length	3.3.9	Steel rule	-	X	1	-	X	2 pair
	Thickness	3.3.9	Micrometer	-	X	1	-	X	
	Wood failure	3.3.9	4.5.3	X	-	1	-	X	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Nylon shank (1 gross pairs)	Mat'l identification	3.3.9.1	1/	-	-	1	-	-	
	Length	3.3.9.1	Steel rule	-	X	1	-	X	2 pair
	Thickness	3.3.9.1	Micrometer	-	X	1	-	X	
Eyelets, 1 pound	Mat'l identification	3.3.11	1/	-	-	-	-	-	
	Thickness	3.3.11	Micrometer	-	X	1	-	X	
	Outside diameter of flange and barrel	3.3.11	Gage	X	-	2	-	X	2 eyelets
	Diameter of hole	3.3.11	Gage	X	-	2	-	X	
	Overall length	3.3.11	Gage	X	-	2	-	X	
	Finish	3.3.11.1	1/	-	-	-	-	-	
Washers, 1 pound	Mat'l identification	3.3.11.2	1/	-	-	-	-	-	
	Thickness	3.3.11.2	Micrometer	-	X	1	-	X	
	Outside diameter of flange	3.3.11.2	Gage	X	-	2	-	X	2 washers
	Diameter of hole	3.3.11.2	Gage	X	-	2	-	X	
	Overall length	3.3.11.2	Gage	X	-	2	-	X	
Steel box toes (pairs)	Mat'l identification	3.3.14	1/	-	-	-	-	-	
	Hardness	3.3.14	E 18 3/	-	-	3	no.	-	2 pair
	Carbon cont.	3.3.14	1/	-	-	-	-	-	
	Coating	3.3.14	1/	-	-	-	-	-	
	Thickness	3.3.14	1/	-	-	-	-	-	
Inner lining sealer	Overall thickness	3.6.1	1/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Nylon inner lining	Overall thickness	3.6.2	1/	-	-	-	-	-	
Inner lining seal strip	Overall thickness	3.6.3	1/	-	-	-	-	-	
Inside friction vamp	Overall thickness	3.6.4	1/	-	-	-	-	-	
Inside toe stiffener	Overall thickness	3.6.5	1/	-	-	-	-	-	
Friction filler	Overall thickness	3.6.7	1/	-	-	-	-	-	
Heel stiffener	Overall thickness	3.6.8	1/	-	-	-	-	-	
Heel stiffener (alternate)	Overall thickness	3.6.9	1/	-	-	-	-	-	
Inside friction counter	Overall thickness	3.6.10	1/	-	-	-	-	-	
Inner vamp lining	Coating compound thickness	3.6.11	1/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Inner quarter lining	Coating compound thickness	3.6.12	1/	-	-	-	-	-	
Interlining	Coating compound thickness	3.6.15	1/	-	-	-	-	-	
Lasting strip	Overall thickness	3.6.16	1/	-	-	-	-	-	
Interlining front and back seam strips	Overall thickness	3.6.16.1	1/	-	-	-	-	-	
Outer vamp lining	Coating compound thickness	3.6.17	1/	-	-	-	-	-	
Outer quarter lining	Coating compound thickness	3.6.18	1/	-	-	-	-	-	
Vamp sealer	Thickness of coatings	3.6.19	1/	-	-	-	-	-	
Quarter sealer	Thickness of coatings	3.6.20	1/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Vamp and quarter joints sealer strip	Thickness and width	3.6.20.1	1/	-	-	-	-	-	
Friction sole form	Overall thickness	3.6.21	1/	-	-	-	-	-	
Combination vamp and chafing piece	Thickness of coatings	3.6.22	1/	-	-	-	-	-	
Upper	Thickness	3.6.23	1/	-	-	-	-	-	
Heel stay	Overall thickness	3.6.24	1/	-	-	-	-	-	
Friction toe foxing	Overall thickness and width	3.6.25	1/	-	-	-	-	-	
Toe foxing	Width and thickness	3.6.26	1/	-	-	-	-	-	
Friction heel piece	Overall thickness	3.6.27	1/	-	-	-	-	-	
Heel piece	Thickness	3.6.28	1/	-	-	-	-	-	

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TABLE XXI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot avg	No. determinations per sample unit	Results reported as: Number to nearest	Results reported as: Pass or fail	Sample unit
Lace chafing piece	Overall thickness	3.6.29	1/	-	-	-	-	-	
Eyelet stay cover	Thickness	3.6.30.1	1/	-	-	-	-	-	
Eyelet stay form	Overall thickness	3.6.30.2	1/	-	-	-	-	-	
Outer eyelet stay	Overall thickness	3.6.30.3	1/	-	-	-	-	-	
Inner eyelet stay	Overall thickness	3.6.30.4	1/	-	-	-	-	-	
Lower hinge stay	Thickness	3.6.30.5	1/	-	-	-	-	-	
Upper hinge stay	Overall thickness	3.6.30.6	1/	-	-	-	-	-	

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirements. For the materials required to be coated with a specific compound, the testing required to establish data on which to form a basis for making a certification may be performed on the coated material prior to the components being cut out.

2/ Refers to FED-STD-191.

3/ Refers to ASTM standard.

4/ Except that the coolant shall be methanol, and the test temperature shall be minus 40°F for rubber to be used in type I boots and minus 65°F for rubber to be used in type II boots.

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4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or services for which in-process inspection has indicated non-conformance.

4.4.2.1 Leakage testing. After vulcanization, each boot produced shall be tested for leakage as specified in 4.5.2. Any test failure shall be cause for rejection of the boot and correction of the affected process.

4.4.2.2 Adhesion of component testing. After vulcanization, one boot shall be selected randomly from each vulcanized lot (batch) produced each day and tested for component adhesion adequacy as specified in 4.5.5. Failure of the test shall require 100 percent examination of the vulcanized lot. Each boot tested during the 100 percent examination that fails the test shall be rejected.

4.4.3 End item critical defect examination of type I, class 3 boots. Prior to performing the end item sampling examination required in 4.4.4, the type I, class 3 boots shall be 100 percent examined for the critical defects listed in table XXII. The 100 percent end item critical defect examination shall become a part of the contractor's inspection system or quality program.

4.4.4 End item visual examination. The end items shall be examined for the defects listed in table XXII. The definitions and figures in MIL-STD-284 shall apply. The lot size shall be expressed in units of one boot. The sample unit shall be one boot and the selection shall be by pairs. For the pairing examination, each pair shall be examined together and each pairing defect found shall be classified as a single defect. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. The finding of one or more critical defects shall be cause for rejection of the lot.

TABLE XXII. End item visual defects

Defect	Classification Critical	Major	Minor	Applicable figure in MIL-STD-284
Pairing				
Not tied together or laced, when required.			201	_
Significant variation in shade or color.			202	_
Not right and left of same type and size.		101		_
Design and size				
Not as specified.		102		_

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TABLE XXII. End item visual defects (cont'd)

Defect	Classification Critical	Major	Minor	Applicable figure in MIL-STD-284
Cleanliness				
Grease, oil, or other form of foreign matter on outside or inside of footwear. (Stains on the inside of the boot caused by aluminum last shall not be scored as defects.)			203	-
Color and finish				
Exterior of footwear, including all components, not color specified.		103		-
Interior of footwear not color specified or not color evidenced in the approved sample.			204	-
Exterior of footwear, including all components, not the finish specified, color not uniform, or significant difference in shade when compared to approved sample.			205	-
Construction and assembly				
NOTE: The critical area is any part of the insulated area.				
Any component missing.		104		-
Any operation omitted or not properly performed, or any part missing.		105		-
Any part malformed:				
Critical area		106		-
Uncritical area			206	-
Poor or incomplete vulcanization 1/.		107		-
Cuts, tears, or holes.		108		4,5,6,19,20
Poor adhesion or looseness (component can be peeled) 1/.		109		7,8,9,10,11
Any bonded joint that has been separated 1/.		110		-
Lumps:				-
Any in critical area		111		
Uncritical area - two or more lumps over 1/16 inch but less than 1/4 inch or one or more lumps 1/4 inch or more.			207	
Any crease		112		-

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TABLE XXII. End item visual defects (cont'd)

Defect	Classification Critical	Major	Minor	Applicable figure in MIL-STD-284
Wrinkled stock:				21,22,23
Any in critical area		113		
Uncritical area - 1/2 inch or more		114		
- up to 1/2 inch			208	
Marred stock:				3,12,13,14,15
Any in critical area		115		
Uncritical area - 1/2 inch or more		116		
- up to 1/2 inch			209	
Burned gum		117		-
Started heel		118		17
Started sole		119		18
Bloom			210	-
Blown stock:				2
Any in critical area		120		
Uncritical area - 1/8 inch or more		121		
- up to 1/8 inch			211	
Blister:				1
Any in critical area		122		
Uncritical area - 1/4 inch or more		123		
- up to 1/4 inch			212	
Dirt: (embedded foreign material)				1
Any in critical area		124		
Uncritical area			213	
Pits:				16
Any in critical area		125		
Uncritical are - 1/8 inch or more		126		
- up to 1/8 inch			214	
Any air pocket in the insulated area.		127		-
Any air pocket other than in the insulated area:				-
- 1/4 inch or more		128		
- under 1/4 inch			215	
Poor or uneven lasting.		129		-
Metal parts				
Poorly attached, poor finish. or not as specified.		130		-
Number of eyelets not as specified but each row has the same number.			216	-
Not the same number of eyelets in each row.		131		-

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TABLE XXII. End item visual defects (cont'd)

Defect	Classification Critical	Major	Minor	Applicable figure in MIL-STD-284
Metal parts (cont'd)				
Eyelets not properly spaced within the row causing misalignment between the rows to the extent of interfering with proper lacing:				-
Seriously affecting serviceability or appearance		132		
Affecting serviceability or appearance but not seriously.			217	
Edge of any eyelet less than 3/16 inch from front edge of boot to the edge of the eyelet.		133		-
Edge of any eyelet less than 1/4 inch but not less than 3/16 inch or more than 1/2 inch from front edge of boot to the edge of eyelet			218	-
Any eyelet not securely clinched.			219	-
Any eyelet washer on outside surface of boot			220	
Any sharp edge or metal sliver.			221	-
Any part missing.		134		-
Steel box toe omitted (type I, class 3 boot) 2/ 1				-
Valve				
Valve missing when required, not specified type, or damaged or otherwise inoperative.		135		-
Valve open.		136		-
Boot form				
Boot form omitted or boot not properly laced to form to achieve proper crease of gusset.			222	-
Marking (all types including instruction tags)				
Instruction tags: Omitted, incorrect, illegible, incomplete, not accomplished in the specified manner, or not in the proper location.		137		-
Not secured to boot as specified			223	-
All other markings: Omitted, incorrect, illegible, incomplete, not accomplished in the specified manner, or not in the proper location.			224	-
ANSI label omitted (type I, class 3 boot)			225	-

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1/ These defects shall be verified by attempting to separate the component layers by hand. Any evidence of separation shall be scored as a defect.

2/ Determine the presence of the steel box toe by trying to depress by hand the toe portion of the boot. If the toe area can be depressed, the steel box toe has been omitted.

4.4.5 End item dimensional examination. The end items shall be examined for the dimensions specified in table XX and 3.6.42. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of one boot. The sample unit shall be one boot and the selection shall be by pairs. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.6 Examination for accuracy of weight marking. The boots shall be examined to determine that the weight of the boot conforms to the weight marking branded in the heel as specified in 3.6.46. Any resultant weight that is not within 0.5 ounce of the branded weight shall be classified as a defect. The lot size shall be expressed in units of one boot. The sample unit shall be one boot without laces, instruction tag, and boot form. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.7 End item testing. The boots shall be tested for leakage as specified in table XXIII. The type I, class 3 boot shall be tested for impact resistance as specified in table XXIII. The lot size for all tests shall be expressed in units of one boot. The sample unit for each test shall be one boot. The inspection level for the leakage test shall be S-4 and for the impact resistance, S-2 . To the extent limited by the sample size, as many sizes as possible that are present in the lot shall be tested. If there are fewer sizes present in the lot than the sample size, a proportional sample shall be selected. Any leakage or impact resistance test failure shall be cause for rejection of the lot. (Note: Testing for leakage is a verification of the in-process testing required in 4.4.2.1.)

TABLE XXIII. End item tests

Characteristic	Specification reference		Number determinations per sample unit	Results reported as
	Requirement	Test method		Pass of Fail
Impact resistance (type I, class 3 only)	3.9.1	4.5.4	1	X
Leakage	3.9.2	4.5.2	1	X

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4.4.8 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Content	Number of pairs of boots per container is more or less than required.

4.4.9 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 Tricot nylon cloth elongation test. Elongation of the tricot nylon cloth shall be determined in accordance with 4.5.1.1 or 4.5.1.2.

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4.5.1.1 Inclined plane type machine. The elongation of the cloth shall be measured on an inclined plane type of machine (i.e., IP 4 or similar type). A suitable autographic recording device shall be provided. Five specimens of the cloth shall be cut 4 inches by 6 inches with the long dimensions parallel to the course direction and each specimen shall be cut so as to contain different wale and course areas. The specimen shall be placed in flat type clamps which shall be 1 inch wide and 3 inches apart. The clamps which shall face each other shall be aligned with the wales of the sample. The specimen shall be placed in the clamp jaws without tension and without slack. Seven 5-pound weights, as used in the IP 4 type machine, shall be placed in the carriage furthest away from the jaws. Each specimen shall be run through ten complete cycles with the machine proceeding to a maximum of 40 pounds tension on each cycle. The maximum elongation of each specimen shall be recorded and the average of the five maximum elongations shall be calculated. The percent elongation shall be calculated as follows:

$$\text{Percent elongation} = \frac{\text{Average elongation - Initial distance between clamps}}{\text{Initial distance between clamps}} \times 100$$

4.5.1.2 Scale pendulum type machine. The elongation of the cloth shall be measured on a 150-pound scale pendulum type machine equipped with a suitable autographic recording device. Specimens of the cloth shall be cut 4 inches by 6 inches with long dimension parallel to the course direction. The specimen shall be placed in flat type clamps which shall be 1 inch wide and 3 inches apart. The clamps which shall face each other shall be aligned with the wales of the sample. The sample shall be placed in the clamp jaws without tension and without slack. The machine shall run through ten complete cycles, recording the elongation on the autographic chart. Five specimens cut from different wale and course areas shall be tested with individual recorded hysteresis curves. During the test, the pawls shall be disengaged and the speed of the pulling clamp shall be 12.0 $\pm$ 0.5 inches per minute. The apparatus shall be allowed to proceed to a point of depression to exert a 40-pound stress, and then this stress shall be unloaded by the reversing lever. The final hysteresis curve shall be completed by turning off the electrical switch of the apparatus at the 40-pound stress level. The final elongation for each specimen shall be obtained from the chart after ten applications of a 40-pound load and shall be measured at the 40-pound stress level of the last recorded cycle. The result reported shall be the average of five specimens tested.

4.5.2 Leakage test. The boot shall be tested for leakage by applying 0.5 pounds per square inch of air pressure through the air valve. The pressurized boot shall then be immersed in water up to within a minimum of 1.5 inches from the boot leg top for a minimum of 30 seconds and observed for leakage. Any steady stream of air bubbles shall be considered evidence of leakage and a test failure.

4.5.3 Wood shank wood failure test. A test specimen shall consist of a complete wood shank unit. A knife blade shall be inserted into the glue lines between the core and the face and back veneers of the test specimen at several points, and shall be worked so as to cause a separation. Each test area shall be examined for percent of wood failure. Any test area that has less than 20 percent of wood failure shall be considered as a test failure.

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4.5.4 Impact resistance test of steel toe, type I, class 3 boot. The impact test shall be performed on the toe section of the boot and in accordance with the following procedure: A falling weight of steel or other suitable material, weighing not less than 49-1/2 pounds nor more than 50-1/2 pounds, 1 (± 0.020) inch in diameter and with a striking face made hemispherical in shape by rounding on a 1 inch radius, equipped to drop freely in a tube or other suitable guide, shall be used. The weight shall be dropped from a distance of 18 (± 1/4) inches above the top of the box toe. As an alternate, a falling weight of 25 (± 1/4) pounds shall be dropped from a distance of 36 (± 1/2) inches. The equipment shall be assembled so that the center of the striking face of the falling weight will strike the boot at midwidth, 1/2 inch in front of the back edge of the box toe. The boot (or toe section) shall be mounted on an anvil (or rigidly mounted bedplate) of steel or other suitable material weighing not less than 500 pounds, equipped with suitable guides (or clamps) to hold the boot specimen in place. In performing the test, the minimum inside clearance of 1/2 inch may be determined instantaneously by placing lumps of moldable material such as wax or modeling clay inside the boot directly below the point of impact. The moldable material shall be compressed with the fingers so that it contacts both upper and lower inside surfaces of the boot before the test is performed. When in place, the greatest horizontal dimension of the lumps shall not exceed 1 inch. An approved method for impact testing is contained in American National Standard ANSI Z 41.1. The test shall be considered a failure if the moldable material is compressed to a point less than 1/2 inch in thickness.

4.5.5 Adhesion test of boot components. The boots shall be tested for adequacy of adhesion of the outside counter by attempting to peel back by hand the exposed edges of the outside counter components. Any evidence of component separation (delamination) or looseness more than 1/2 inch long by 1/8 inch deep shall be considered a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each boot shall have a non-bending grade paperboard stay inserted the full height of the uppers. The paperboard stay shall be 0.032 inch minimum thickness and shall measure approximately 5 inches by 11-1/2 inches.

5.1.2 Commercial. The boots shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each pair of boots of one type ans size only, preserved as specified in 5.1, shall be separated one from the other by a sheet of kraft paper not less than 30 pounds basic weight (24 by 36), conforming to A-A-203. Each pair of boots shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. The approximate dimensions of each shipping container shall be as specified in table XXIV. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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TABLE XXIV. Shipping container dimensions

Range of boot sizes	Length inches	Width inches	Depth inches	Pallet number
3 - 6 (all types and widths)	18-1/2	11	4-1/2	8
7 - 10 (all types and widths)	20	12	5-1/4	102
11 - 14 (all types and widths)	21-1/2	13	6-1/4	6

5.2.2 Level B packing. Each pair of boots of one type and size only, preserved as specified in 5.1, shall be separated one from the other by a sheet of kraft paper not less than 30 pounds basic weight (24 by 36), conforming to A-A-203. Each pair of boots shall be packed in a fiberboard shipping container conforming to type CF, domestic class, grade 275, style RSC of PPP-B-636. The approximate dimensions of each shipping container shall be as specified in table XXIV. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping containers shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. Boots preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), boots, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F and G. Pallet pattern numbers shall be as specified in table XXIV in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boots are intended for use in wet-cold, dry-extreme cold climates by the personnel of the Department of Defense as indicated below. The type I boot will provide protection in temperatures below -20°F and above.

Type I class 1 - Army, Air Force, and Marine Corps

Type I, classes 2 and 3 - Navy

Type II, class 2 - Army, Air Force, and Marine Corps

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. Sizes and widths required (see 1.2).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is required (see 3.2, 4.3, and 6.3).
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Type and class of unit load (see 5.2.1).
h. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
i. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Master models and patterns. Models of the lasts, last templates and last patterns in the following sizes and widths will be loaned by the Government to the contractor upon request to the contracting activity. Master model (right foot) last and leg (wood) 5, 7, 9, 11, 13; widths: extra narrow, narrow, regular, wide, extra wide. Heel curve templates (finishing) paper size 4 through 14; widths: extra narrow, narrow, regular, wide, extra wide. Back heel curve templates, 10 inches high, paper sizes 4 through 14; widths: extra narrow, narrow, regular, wide, extra wide. Toe templates paper, sizes 4 through 14; widths: extra narrow, narrow, regular, wide, extra wide. Templates (heel curve, back heel curve and toe) for size 3 may be graded down from the size 4.

6.6 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG 2333. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Valve. Sources of supply for the valve specified in 3.3.12 is:

Schrader Automotive Products
(Division of Scovill)
Dickson, TN 37055

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6.8 Material sources. The special materials described in this specification may be obtained from the following sources. There may be equivalent items available from other suppliers.

Insulating foam	Delsolite N754	Nylco Corp.
Expanded polyurethane	Bayflex 440	Mobay Corp.
Outsole polyurethane	Bayflex 408	Mobay Corp.

6.9 Supersession data. Type II, class 1 boot and its requirements have been deleted since it is no longer required in the system.

6.10 Subject term (key word) listing.

Footwear
Safety toe

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force 99

Review activities:

Army - MD
DLA - CT

User activities:

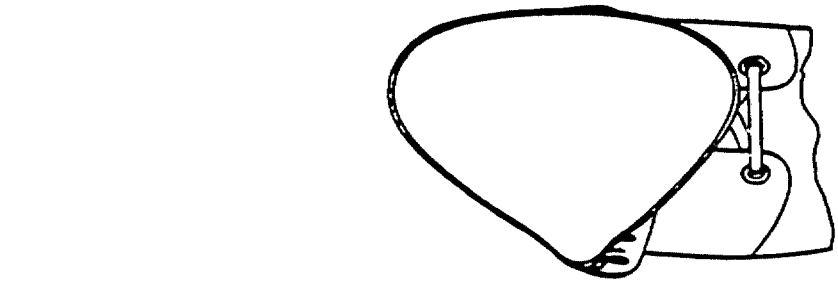
Navy - MC
Air Force - 11

Preparing activity:

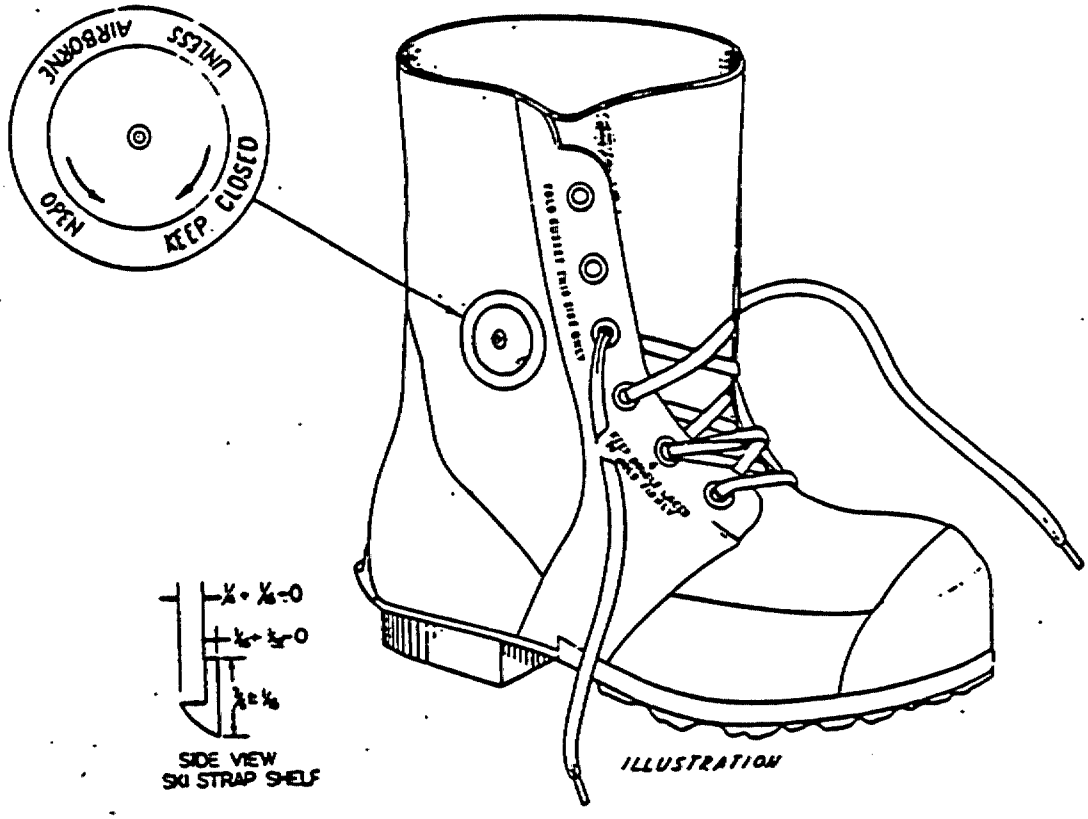
Army - GL

(Project 8430-0387)

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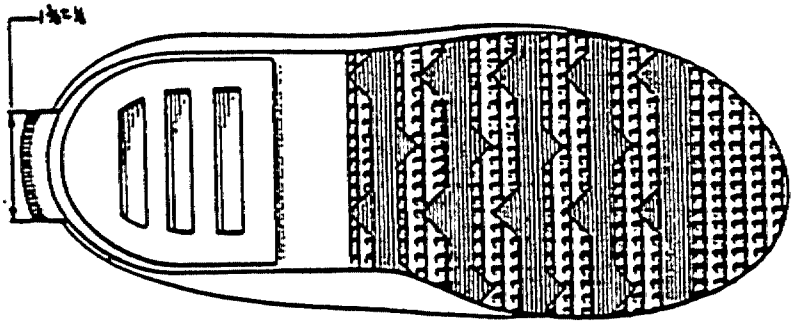


TOP



ILLUSTRATION

SIDE VIEW
SKI STRAP SHELF



BOTTOM

FIGURE 1 TYPE I, CLASS 1

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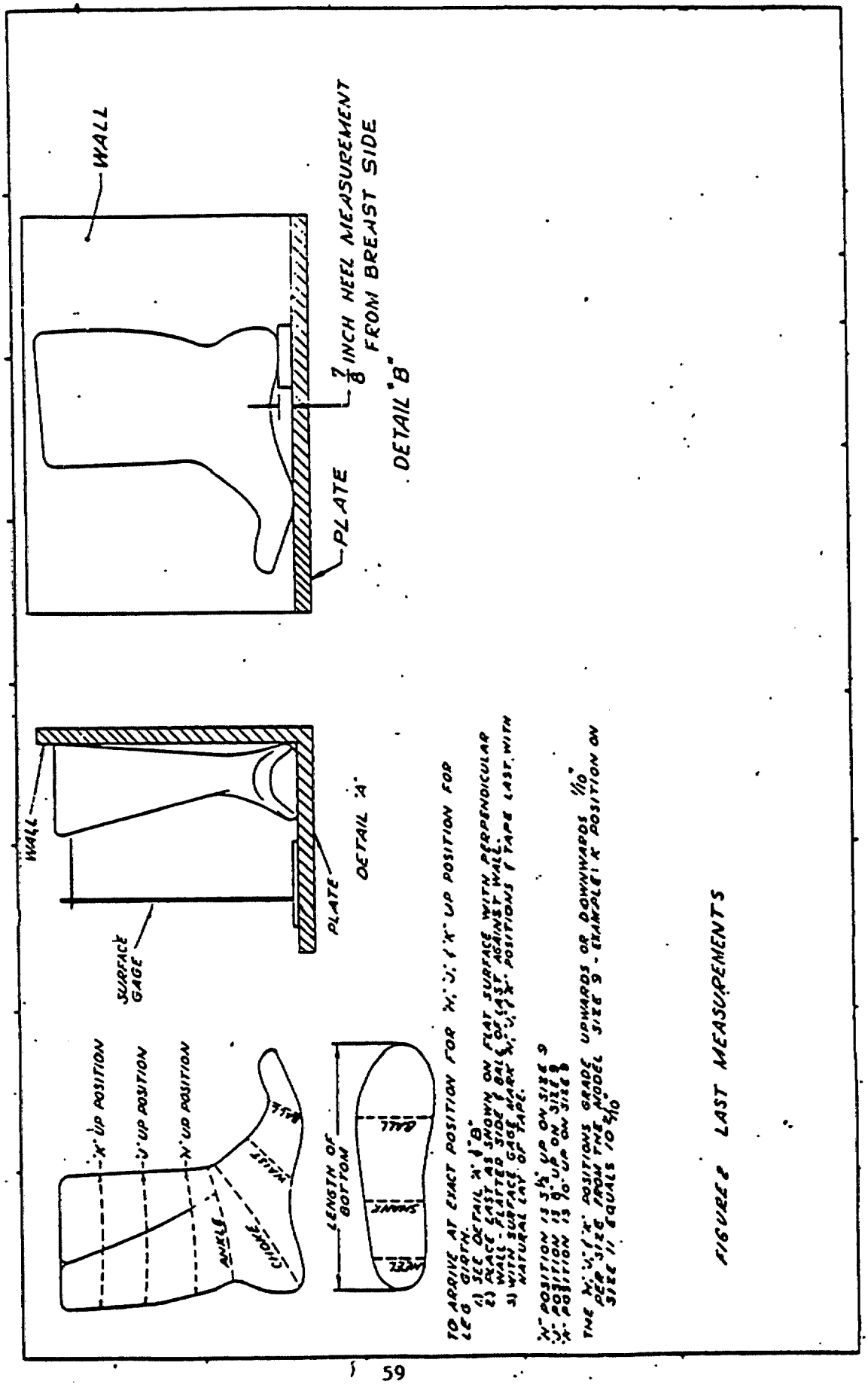
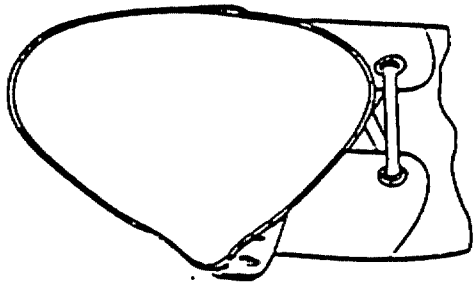
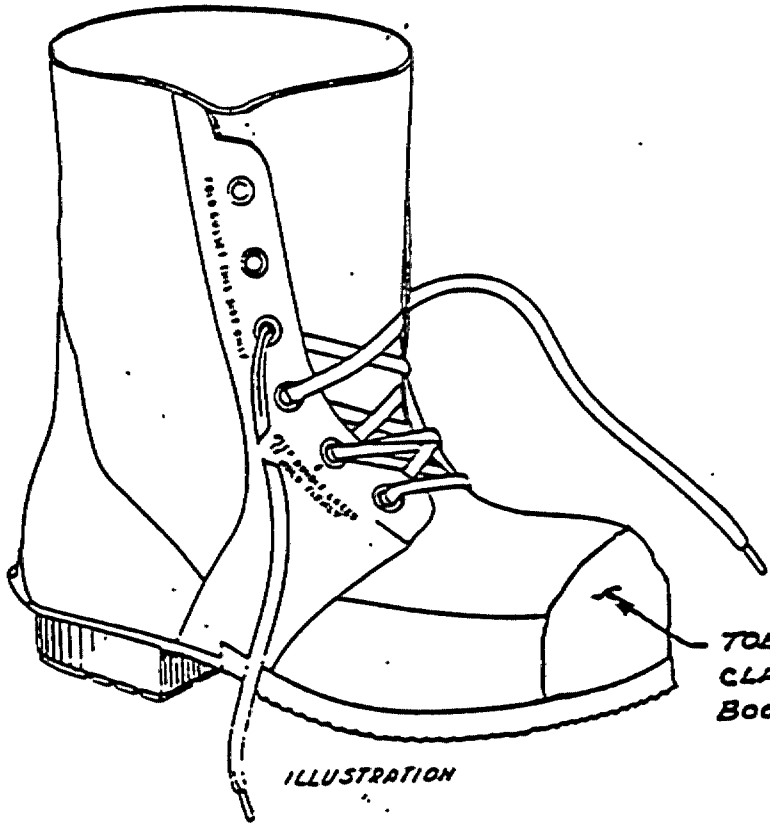


FIGURE 8 LAST MEASUREMENTS

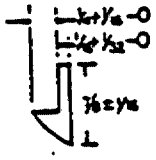
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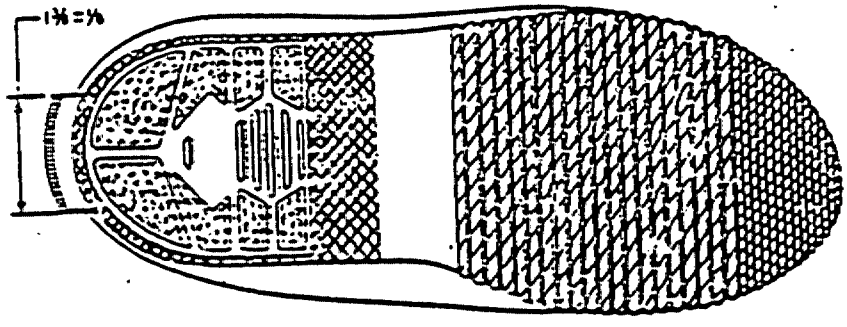
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ILLUSTRATION



SIDE VIEW
SKI STRAP SHELF



BOTTOM

FIGURE 3 TYPE I, CLASS 2 AND CLASS 3

2-1-457 E

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Figure 4 - Instruction tag for type I boot

INSTRUCTION TAG

BOOTS, COLD WEATHER, INSULATED, RUBBER (BLACK) FOR WET-COLD USE

TYPE I, CLASS 1, RIGID SOLE

TYPE I, CLASS 2, FLEXIBLE SOLE
TYPE I, CLASS 3, FLEXIBLE SOLE WITH SAFETY TOE

FACTS ABOUT THE INSULATED BOOT

These boots have been designed to protect your feet from cold injury and frost bite in areas where moisture and cold are critical factors, and where the mean monthly temperature ranges between 14°F and 68°F. The boot should not be worn where temperatures fall below - 20°F. (Note: For temperatures below - 20°F, the type II, class 1 boot should be worn.)

The part of the boot that keeps the foot warm consists of layers of wool and felt that are sealed between the rubber inner and outer layers of the boot so that no water can get into the insulation areas.

Marching, running, or heavy work for long periods will cause the feet to sweat. However, this moisture cannot damage the insulation because of the water-proof rubber lining inside the boot. The feet, even if damp, will stay warm. After a few days, your feet become used to this feeling. If these boots are worn continuously, the skin appears wrinkled as if it had been soaked in water. Drying the feet and putting on dry clean socks will cause this condition to disappear. If ice water spills into the boot, it warms to body temperature rapidly.

FOLLOW THESE INSTRUCTIONS:

Wear with one pair of Socks, Wool, Cushion Sole. Fold the top of the sock over the top of boot to prevent slipping and wrinkling. Field trousers are to be worn outside the boots.

SIZES: Boots are supplied in whole sizes. Widths are EXTRA NARROW, NARROW, REGULAR, WIDE, EXTRA WIDE. Try on the same size insulated boots as your leather boot. If you wear a half size in leather boots, try on the next larger or smaller size for proper fit. MAKE SURE THE INSULATED BOOTS FIT PROPERLY. They have a "soft" fit. Be sure yours are large enough. Fit snug, but not tight.

The boot is double laced through the second eyelets. Loosen laces and draw boot onto foot, seating the back of the heel firmly into the boot. Pull laces right above the second eyelet firmly to anchor the boot to the foot. Lace rest of boot comfortably. Fold or tuck fullness of gusset rearward on outerside of leg, per instructions imprinted on the boot. When doffing boot, loosen laces above second eyelet, then complete by loosening laces directly below second eyelet.

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Figure 4 - Instruction tag for type I boot (cont'd)

Change socks at least once daily.

If your feet begin to get cold, button up all your garments and exercise the entire body by swinging the arms, stamping the feet, and rapid movement of the toes.

Wash the inside of boots with soapy water once a month.

Boots should be inspected after each use for holes, cuts and punctures as the insulation can become wet if the boot is accidentally punctured. Repair all outside and inside punctures immediately to prevent moisture from wetting the sealed insulation, using the "Emergency Individual Repair Patch Kit for Insulated Boots." If this kit is not available, temporary repairs can be made by patching with rubber or friction tape, pressure sensitive adhesive cellophane tape or even chewing gum. Damaged boots should be turned in for repair or for new boots.

The shelf on the heel holds a ski or snowshoe binding in place.

For type I class 1, 2 and 3, the air release valve on the outside of the boot equalizes air pressure at different altitudes. To reduce boot swelling and foot discomfort when the boot is worn in unpressurized aircraft or at high altitudes, open the valve allowing pressure to escape. Back at lower altitudes, close again. Be sure to keep closed, except while airborne, to eliminate moisture penetration. DO NOT TAMPER WITH AIR VALVE. USE ONLY WHEN NEEDED!

Do not paint the inside of the boot for identification purposes or polish the boots with shoe polish as this will result in deterioration of the rubber.

CLEANING:

Sprinkle lightly with general purpose scouring powder or similar cleaning powder. Wet a soap impregnated steel wool pad and rub lightly. Rinse with water.

TRENCH FOOT AND FROSTBITE are serious cold injuries which can cuase painful and permanent disability. These injuries can be prevented by proper care of your feet. No boot will keep your feet from freezing if in sub-zero weather you remain motionless for several hours. Wear the insulated boot properly and DO NOT BECOME A COLD WEATHER CASUALTY.

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Figure 5 - Instruction tag for type II boot

INSTRUCTION TAG

BOOTS, COLD WEATHER, INSULATED, RUBBER (WHITE) FOR DRY-EXTREME COLD USE

TYPE II, CLASS 1

FACTS ABOUT THE INSULATED BOOT

These boots have been designed to protect you in sub-zero conditions and must be worn when actual tempratures of - 20°F or below may occur.

The part of the boot that keeps the foot warm consists of layers of wool and felt that are sealed between the rubber inner and outer layers of the boot so that no water can get into the insulation areas.

Marching, running, or heavy work for long periods will cause the feet to sweat. However, this moisture cannot damage the insulation because of the waterproof rubber lining inside the boot. The feet, even if damp, will stay warm. After a few days, your feet become used to this feeling. If these boots are worn continuously, the skin appears wrinkled as if it had been soaked in water. Drying the feet and putting on dry clean socks will cause this condition to disappear. If ice water spills into the boot, it warms to body temperature rapidly.

FOLLOW THESE INSTRUCTIONS:

Wear with one pair of Socks, Wool, Cushion Sole. Fold the top of the sock over top of boot to prevent slipping and wrinkling. Field trousers are to be worn outside the boots.

SIZES: Boots are supplied in whole sizes. Widths are EXTRA NARROW, NARROW, REGULAR, WIDE, EXTRA WIDE. Try on the same size insulated boots as your leather boot. If you wear a half size in leather boots, try on the next larger or smaller size for proper fit. Make sure the insulated boots fit properly! They have a "soft" fit. Be sure yours are large enough. Fit snug, but not tight.

The boot is double laced through the second eyelets. Loosen laces and draw boot onto foot, seating the back of the heel firmly into the boot. Pull laces right above the second eyelet firmly to anchor the boot to the foot. Lace rest of boot comfortably. Fold or tuck fullness of gusset rearward on outerside of leg per instructions imprinted on the boot. When doffing boot, loosen laces above second eyelet, then complete by loosening lace directly below second eyelet.

Change socks at least once daily.

If your feet begin to get cold, button up all your garments and exercise the entire body by swinging the arms, stamping the feet, and rapid movement of the toes.

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Figure 5 - Instruction tag for type II boot (cont'd)

Wash the inside of the boots with soapy water once a month.

Boots should be inspected after each use for holes, cuts and punctures as the insulation can become wet if the boot is accidentally punctured. Repair all outside and inside punctures immediately to prevent moisture from wetting the sealed insulation, using the "Emergency Individual Repair Patch Kit for Insulated Boots." If this kit is not available, temporary repairs can be made by patching with rubber or friction tapes, pressure sensitive adhesive cellophane tape or even chewing gum. Damaged boots should be turned in for repair or for new boots.

The shelf on the heel holds a ski or snowshoe binding in place.

The air release valve on the outside of the boot equalizes air pressure at different altitudes. To reduce boot swelling and foot discomfort when the boot is worn in unpressurized aircraft or at high altitudes, open the valve allowing pressure to escape. Back at lower altitudes, close again. Be sure to keep closed, except while airborne, to eliminate moisture penetration. DO NOT TAMPER WITH AIR VALVE. USE ONLY WHEN NEEDED!

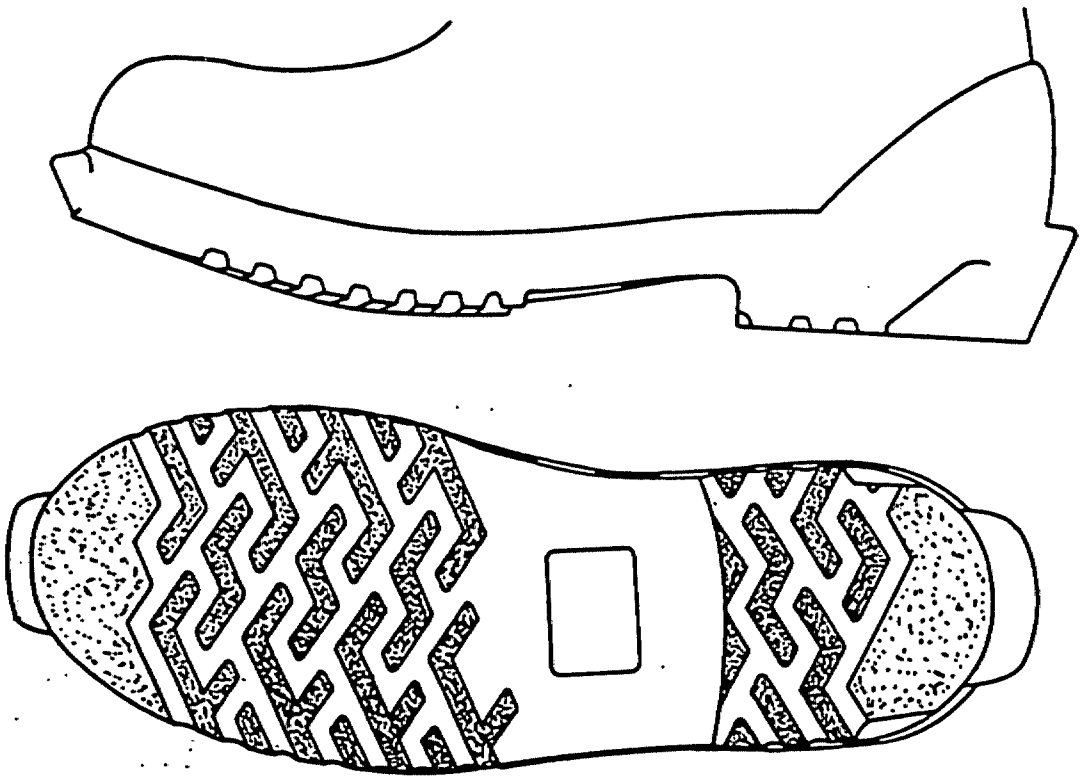
Do not paint the inside of the boot for identification purposes or polish the boots with shoe polish as this will result in deterioration of the rubber.

CLEANING:

Sprinkle lightly with general purpose scouring powder or similar cleaning powder. Wet a soap impregnated steel wool pad and rub lightly. Rinse with water.

TRENCH FOOT AND FROSTBITE are serious cold injuries which can cuase painful and permanent disability. These injuries can be prevented by proper care of your feet. No boot will keep your feet from freezing if in sub-zero weather you remain motionless for several hours. Wear the insulated boot properly and DO NOT BECOME A COLD WEATHER CASUALTY.

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ILLUSTRATION

FIGURE 6, TYPE II, CLASS 2

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